Money and Success in Football

Does Financial Strength Predict League Position? *Sports Economics Seminar — Summer Semester 2026*

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Contents

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1 Executive Summary

Scale of the study: 975 team-seasons across 5 major European leagues (Bundesliga, La Liga, Ligue 1, Premier League, Serie A) covering 10 seasons (2015–16 to 2024–25). All leagues and seasons included in the analysis.

Core finding: The Pearson correlation between Financial Rank and final league position is **r = 0.813** (Spearman = 0.814, both $p < 0.001$). The best-fitting OLS model — Model H (quadratic in Financial Rank) — explains $\mathbf{R^2 = 0.671}$ of the variation in final standings (AIC = 5070.5). Money is the single strongest predictor of league success in European football.

Champions by financial rank: 76% of the 50 league titles in the study were won by the squad with the **highest market value** in their league that season (FR = 1). Only 4 champions came from outside the financial top three — and only one (Leicester City 2015–16) exceeded the ±1.5 SD outlier threshold.

The one true exception: Leicester City 2015–16 (Financial Rank = 11, Premier League) has a Model H residual of **-10.5 positions** — the largest single overperformance in the entire dataset. The model predicted 11th place; they won the title.

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2 Introduction

2.1 Research Context

The five major European football leagues collectively generate over €18 billion in annual revenue, with squad transfer spending exceeding €12 billion in recent seasons. A long-standing debate in sports economics asks: **does financial investment in squad quality causally determine sporting success, or can well-organised, under-funded clubs systematically compete?**

This question matters beyond football. It has implications for competitive balance policy (UEFA Financial Fair Play), fan engagement, and the broader relationship between resource endowments and tournament outcomes.

2.2 Research Questions

This study addresses three interrelated questions using ten years of data from five top-tier European leagues:

1. **Direction and magnitude:** Is there a statistically significant relationship between a squad's financial rank within its league and its final position?
2. **Model form:** Does a linear or quadratic specification better capture the relationship? Do the effects differ across leagues or time?
3. **Exceptions:** Which clubs defy the financial hierarchy, and how statistically unusual are they?

2.3 Contribution

Prior work has established a positive link between payroll/market value and performance (Szymanski & Kuypers, 1999; Hall, Szymanski & Zimbalist, 2002; Frick, 2007). This study contributes a **within-league rank normalisation** that removes between-league scale differences, enabling a pooled analysis across leagues with very different absolute spending levels. It also introduces a systematic residual-based framework to identify and quantify genuine statistical outliers.

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3 Data & Variables

3.1 Dataset Overview

```
desc_raw <- read_csv("outputs/01_descriptive_statistics.csv",
                     show_col_types = FALSE)

desc_raw |>
  select(Group, N, Mean_Position, SD_Position,
         Mean_MV_M_EUR, SD_MV_M_EUR,
         Corr_FR_Pos, Spearman_FR_Pos) |>
  mutate(across(where(is.numeric), ~ round(., 2))) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Group", "N", "Mean Pos.", "SD Pos.",
                  "Mean MV (M€)", "SD MV (M€)",
                  "Pearson r", "Spearman  "),
```

```r
    caption   = "Table 1. Descriptive statistics by league and combined."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(1, bold = TRUE, background = "#EBF5FB") |>
  column_spec(7:8, bold = TRUE, color = "#196F3D")
```

Table 1: Table 1. Descriptive statistics by league and combined.

Group	N	Mean Pos.	SD Pos.	Mean MV (M€)	SD MV (M€)	Pearson r
ALL LEAGUES (combined)	**975**	**10.27**	**5.66**	**277.9**	**254.4**	**0.81**
Bundesliga	180	9.50	5.20	237.2	205.6	**0.78**
La Liga	200	10.50	5.78	256.7	269.4	**0.79**
Ligue 1	196	10.32	5.69	184.7	193.9	**0.79**
Premier League	200	10.50	5.78	458.0	303.5	**0.80**
Serie A	199	10.47	5.78	246.7	182.4	**0.89**

All 975 observations are complete (no missing values on Financial Rank, League Position, or Market Value). The dataset covers 5 leagues × 10 seasons each.

3.2 Variable Definitions

Table 2: Table 2. Variable definitions.

Variable	Type	Range	Description
`League_Position`	Integer	1–18/20	Final league standing; 1 = champion, higher = worse
`Financial_Rank (FR)`	Integer	1–18/20	Squad rank by market value within league-season; 1 = richest
`Market_Value_M_EUR`	Numeric	10–3300+	Total squad market value in millions of euros (Transfermarkt)
`Normalized_Value`	Numeric	0–1	Market value normalised within league-season (min-max or share)
`Log_Normalized_Value`	Numeric	$-\infty$–0	Natural log of Normalized_Value; linearises the skewed MV distribution
`Champion_bin`	Binary (0/1)	0 or 1	1 if the team won the league title that season, 0 otherwise
`Season_Index`	Integer	1–10	Numeric season counter: 1 = 2015–16, 10 = 2024–25

Key design choice — Financial Rank vs. raw market value: Using FR instead of the raw market value removes between-league scale differences. A squad with €300M value is the *richest* in Ligue 1 but *fifth-richest* in the Premier League. FR normalises these differences, allowing pooled analysis.

3.3 Distributions

```r
ggplot(df, aes(x = Financial_Rank, fill = League)) +
  geom_histogram(binwidth = 1, color = "white", alpha = 0.85) +
  facet_wrap(~ League, scales = "free_y", ncol = 3) +
  scale_fill_manual(values = league_colours, guide = "none") +
  scale_x_continuous(breaks = seq(1, 20, 3)) +
  labs(
    x        = "Financial Rank (1 = richest squad)",
    y        = "Count",
    title    = "Distribution of Financial Rank by League",
    subtitle= sprintf("N = %s team-seasons; values near-uniform within each league",
                      format(n_obs, big.mark = ","))
  ) +
  theme_bw(base_size = 12) +
  theme(plot.title = element_text(face = "bold", color = "#1A2A4A"))
```

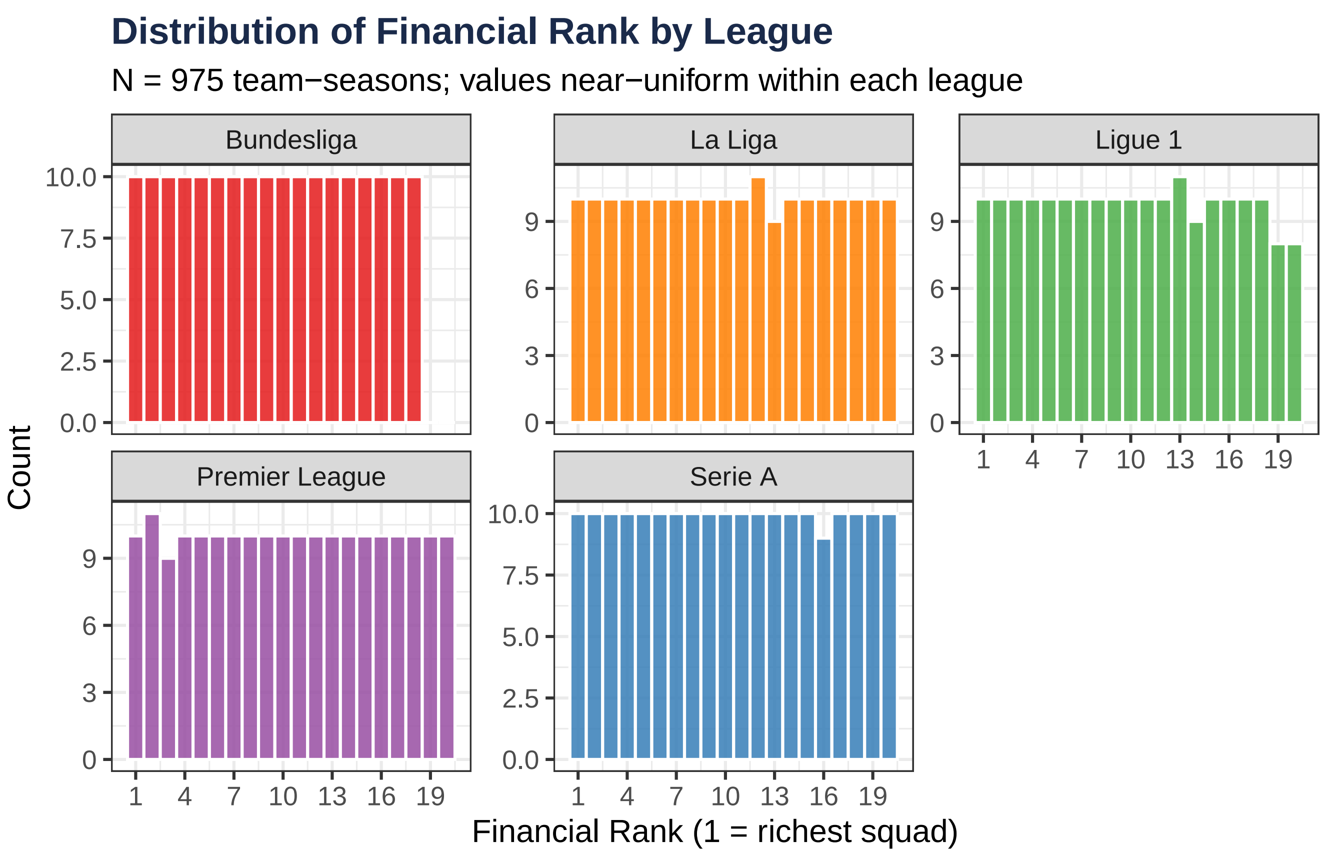

Figure 1: Figure 1. Distribution of Financial Rank across all leagues (2015–16 to 2024–25). Each rank value should appear approximately equally often within a league-season.

```r
df |>
  select(League, Market_Value_M_EUR, Log_Normalized_Value) |>
  pivot_longer(cols = c(Market_Value_M_EUR, Log_Normalized_Value),
               names_to = "Measure", values_to = "Value") |>
  mutate(
    Measure = recode(Measure,
```

```r
    "Market_Value_M_EUR"    = "Raw Market Value (M€)",
    "Log_Normalized_Value" = "Log(Normalized Value)"
  )
) |>
ggplot(aes(x = Value, fill = League)) +
geom_histogram(bins = 30, color = "white", alpha = 0.8) +
facet_grid(Measure ~ League, scales = "free") +
scale_fill_manual(values = league_colours, guide = "none") +
labs(
  x     = "Value",
  y     = "Count",
  title = "Market Value Distributions: Raw vs. Log-Normalised"
) +
theme_bw(base_size = 11) +
theme(plot.title = element_text(face = "bold", color = "#1A2A4A"),
      strip.background = element_rect(fill = "#1A2A4A"),
      strip.text = element_text(color = "white"))
```

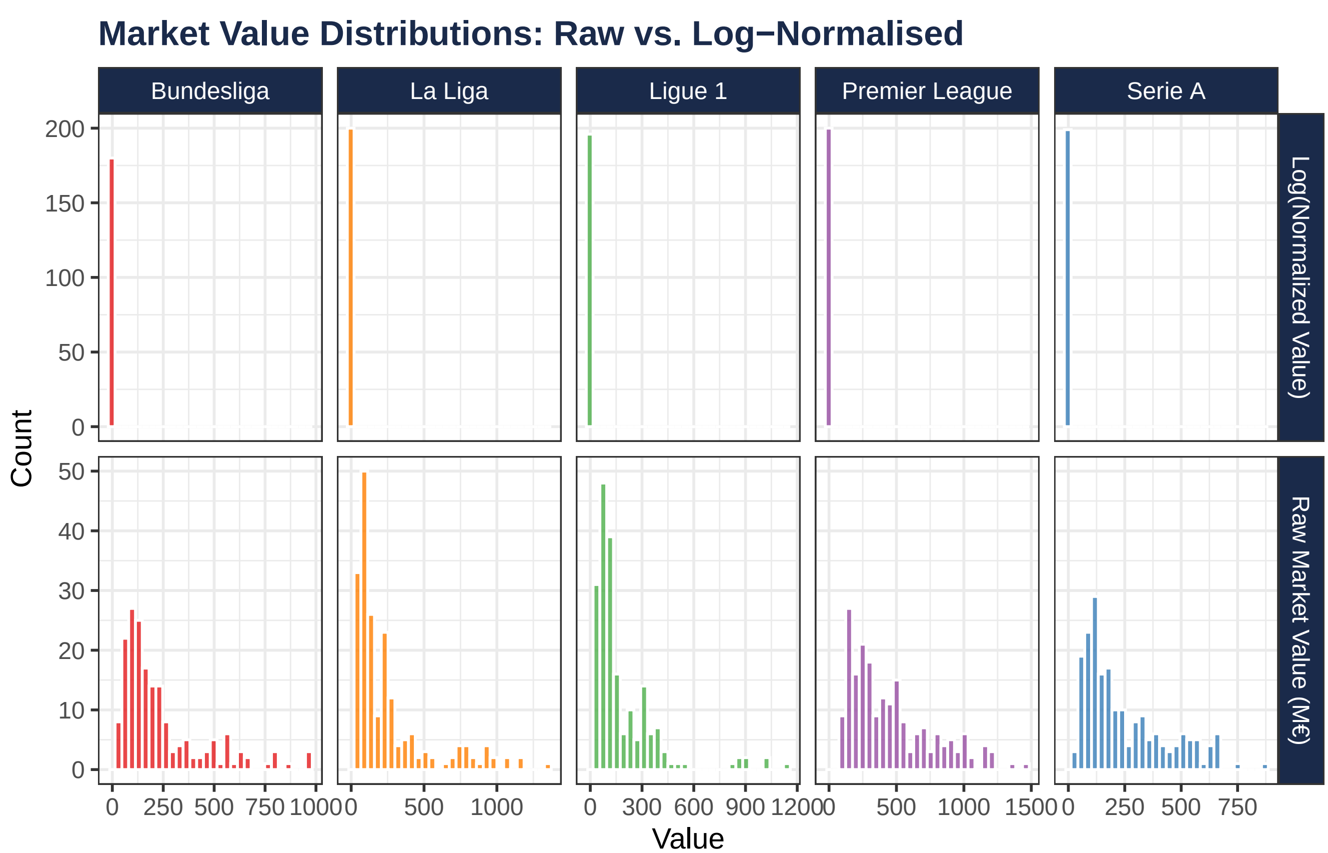

Figure 2: Figure 2. Market value distributions are right-skewed within each league; log-transformation brings values closer to symmetric.

4 Exploratory Analysis

4.1 Financial Rank vs. League Position

```r
# Prediction ribbon for Model H
fr_seq    <- seq(1, max(df$Financial_Rank, na.rm = TRUE), length.out = 200)
pred_main <- predict(
  modH,
  newdata  = data.frame(Financial_Rank = fr_seq),
  interval = "confidence", level = 0.95
) |>
  as.data.frame() |>
  mutate(Financial_Rank = fr_seq)

ggplot(df, aes(x = Financial_Rank, y = League_Position)) +
  geom_ribbon(
    data = pred_main,
    aes(x = Financial_Rank, ymin = lwr, ymax = upr),
    fill = "#1A2A4A", alpha = 0.12, inherit.aes = FALSE
  ) +
  geom_jitter(
    aes(color = League),
    alpha = 0.35, size = 1.8, width = 0.2, height = 0.2
  ) +
  geom_line(
    data = pred_main,
    aes(x = Financial_Rank, y = fit),
    color = "#1A2A4A", linewidth = 1.4, inherit.aes = FALSE
  ) +
  scale_y_reverse(
    breaks = c(1, 5, 10, 15, 20),
    labels = c("1st\n(Champion)", "5th", "10th", "15th", "20th")
  ) +
  scale_x_continuous(breaks = seq(1, 20, 2)) +
  scale_color_manual(values = league_colours) +
  labs(
    x        = "Financial Rank (1 = richest squad)",
    y        = "Final League Position",
    color    = "League",
    title    = "Financial Rank Predicts Final League Position",
    subtitle = sprintf(
      "N = %s team-seasons | Pearson r = %s | Model H R² = %s (quadratic fit shown)",
      format(n_obs, big.mark = ","), r_pearson, r2_H
    )
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", color = "#1A2A4A"),
    plot.subtitle = element_text(color = "#566573"),
    legend.position  = "bottom",
    panel.grid.minor = element_blank()
  )
```

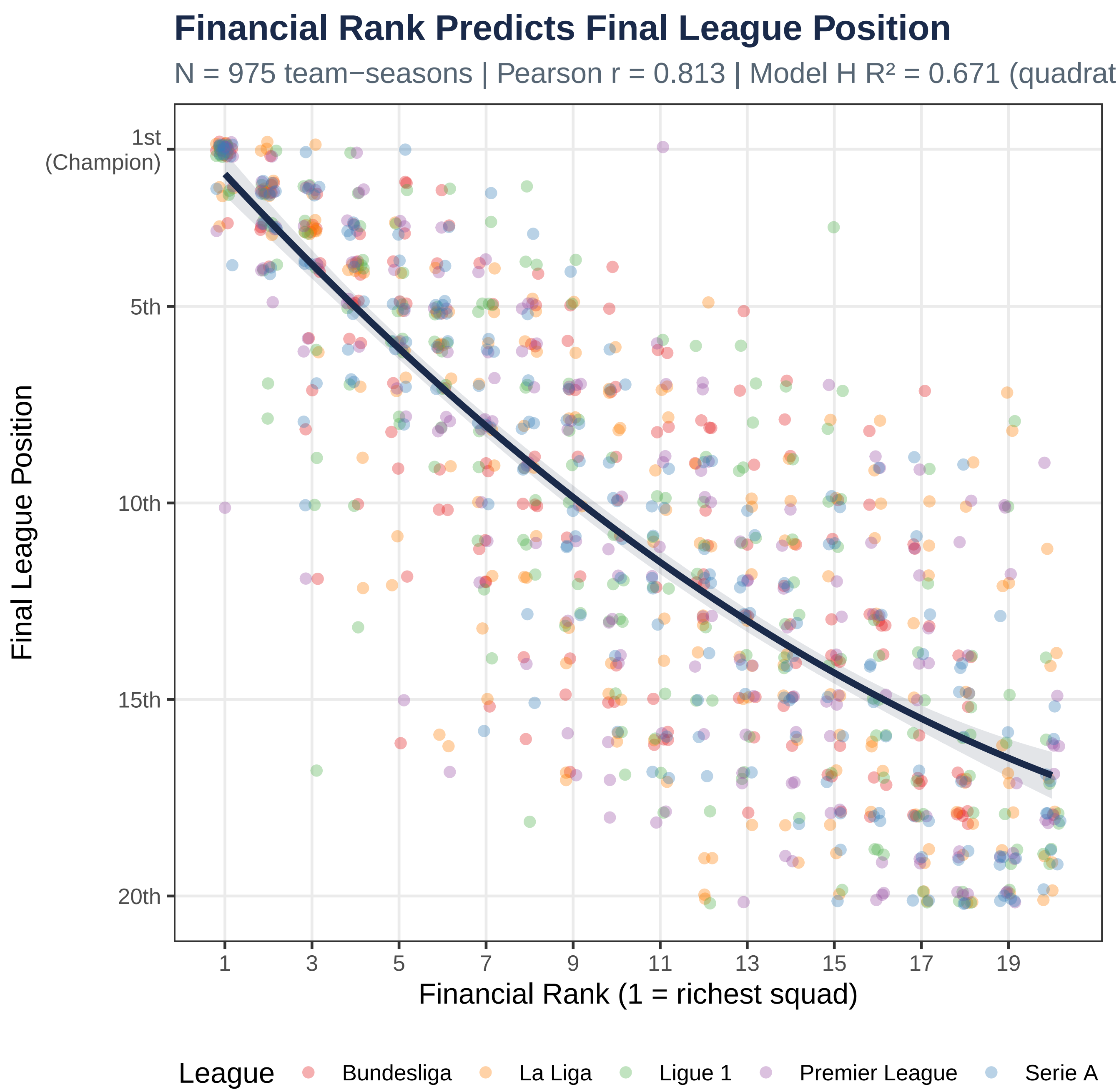

Figure 3: Figure 3. Financial Rank vs. Final League Position for all 975 team-seasons. The quadratic Model H fit (dark line) with 95% confidence band is overlaid. Each colour represents a league.

The relationship is clear and consistent across all five leagues: wealthier squads (lower Financial Rank) finish higher. The slight curvature in the fitted line indicates **diminishing returns** — the positional advantage of being slightly richer is larger at the top of the distribution than at the bottom.

4.2 Correlation by League

```r
league_corr <- read_csv("outputs/10_league_regression_breakdown.csv",
                        show_col_types = FALSE) |>
  mutate(League = factor(League, levels = league_order))

ggplot(league_corr,
       aes(x = fct_reorder(League, Pearson_FR_Pos), y = Pearson_FR_Pos,
           fill = League)) +
  geom_col(alpha = 0.85, width = 0.65) +
  geom_text(aes(label = sprintf("r = %.3f", Pearson_FR_Pos)),
            hjust = -0.15, fontface = "bold", size = 4) +
  geom_hline(yintercept = r_pearson, linetype = "dashed",
             color = "#1A2A4A", alpha = 0.6) +
  annotate("text", x = 0.6, y = r_pearson + 0.005,
           label = sprintf("Combined: r = %.3f", r_pearson),
           hjust = 0, size = 3.5, color = "#1A2A4A") +
  coord_flip(xlim = c(0.5, 5.5), ylim = c(0, 1.05)) +
  scale_fill_manual(values = league_colours, guide = "none") +
  labs(
    x       = NULL,
    y       = "Pearson r (Financial Rank vs. League Position)",
    title   = "FR-Position Correlation Is Strongest in Serie A",
    subtitle= "All correlations significant at p < 0.001"
  ) +
  theme_bw(base_size = 13) +
  theme(plot.title = element_text(face = "bold", color = "#1A2A4A"))
```

```r
league_corr |>
  select(League, N, Pearson_FR_Pos, Spearman_FR_Pos, R2_FR, Coef_FR) |>
  mutate(across(where(is.numeric), ~ round(., 3))) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("League", "N", "Pearson r", "Spearman  ",
                  "R² (FR)", " (FR)"),
    caption   = "Table 3. League-level correlation and simple regression results."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(which(league_corr$League == "Serie A"), bold = TRUE,
           background = "#FEF9E7")
```

Table 3: Table 3. League-level correlation and simple regression results.

League	N	Pearson r	Spearman	R^2 (FR)	(FR)
Bundesliga	180	0.783	0.783	0.613	0.783

Table 3: Table 3. League-level correlation and simple regression results. *(continued)*

League	N	Pearson r	Spearman	R^2 (FR)	(FR)
La Liga	200	0.787	0.787	0.619	0.787
Ligue 1	196	0.791	0.791	0.625	0.791
Premier League	200	0.801	0.801	0.641	0.800
Serie A	**199**	**0.889**	**0.890**	**0.791**	**0.889**

Serie A shows the strongest relationship (r = 0.889, R^2 = 0.791), suggesting that in Italian football the financial hierarchy is particularly rigid. The Premier League (r = 0.801) follows closely, consistent with its reputation for large-payroll clubs dominating.

4.3 Scatter by League

```r
ggplot(df, aes(x = Financial_Rank, y = League_Position, color = League)) +
  geom_jitter(alpha = 0.35, size = 1.4, width = 0.2, height = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x,
              color = "#1A2A4A", fill = "#F5A623",
              se = TRUE, linewidth = 1.1, alpha = 0.2) +
  facet_wrap(~ League, ncol = 3, scales = "fixed") +
  scale_y_reverse(breaks = c(1, 5, 10, 15, 20)) +
  scale_x_continuous(breaks = seq(1, 20, 4)) +
  scale_color_manual(values = league_colours, guide = "none") +
  labs(
    x       = "Financial Rank (1 = richest)",
    y       = "Final League Position",
    title   = "FR-Position Relationship Is Consistent Across All Leagues"
  ) +
  theme_bw(base_size = 12) +
  theme(
    plot.title      = element_text(face = "bold", color = "#1A2A4A"),
    strip.background = element_rect(fill = "#1A2A4A"),
    strip.text       = element_text(color = "white", face = "bold")
  )
```

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5 OLS Regression Analysis

5.1 Model Building Strategy

We estimate a series of nested OLS models, progressively adding complexity:

Step	Question	Model
Baseline	Does FR alone predict position?	A: `Pos ~ FR`
League FE	Do league-level intercepts improve fit?	B: `Pos ~ FR + League`
Alternative predictor	Is log(MV) better than FR?	C: `Pos ~ Log(MV)`

Step	Question	Model
Full alternative	Log(MV) + controls	D: `Pos ~ Log(MV) + League + Season`
Non-linearity	Does a quadratic term help?	**H:** **`Pos ~ FR + FR²`**

Model selection is based on AIC (lower = better) and adjusted R^2. Nested model comparisons use F-tests.

5.2 Model Comparison

```r
# Compute fit statistics for all OLS models using broom::glance()
modG <- lm(League_Position ~ Financial_Rank * League, data = df)

ols_fits <- list(A = modA, B = modB, C = modC, D = modD, G = modG, H = modH) |>
  map_dfr(glance, .id = "Model") |>
  transmute(
    Model,
    Formula = c("FR", "FR + League FE", "Log(MV)",
                "Log(MV) + League + Season",
                "FR × League (interaction)", "FR + FR²  "),
    R2      = round(r.squared, 4),
    Adj_R2  = round(adj.r.squared, 4),
    AIC     = round(AIC, 1),
    BIC     = round(BIC, 1)
  )

# Add logit McFadden R² from pre-computed CSV
logit_row <- read_csv("outputs/09_model_fit_summary.csv", show_col_types=FALSE) |>
  filter(Model == "E") |>
  transmute(Model, Formula = "Champion ~ FR (Logit)",
            R2 = round(R2_McFadden, 4), Adj_R2 = NA_real_,
            AIC = round(AIC, 1), BIC = NA_real_)

bind_rows(ols_fits, logit_row) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Model", "Formula", "R²", "Adj. R²", "AIC", "BIC"),
    caption   = "Table 4. Model comparison.   = preferred specification (lowest AIC among OLS models).",
    align     = "llrrrr"
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(which(bind_rows(ols_fits, logit_row)$Model == "H"),
           bold = TRUE, background = "#FEF9E7") |>
  footnote(
    general = "OLS response: League_Position (1 = champion). R² for Model E is McFadden pseudo-R².",
    general_title = "Note:"
  )
```

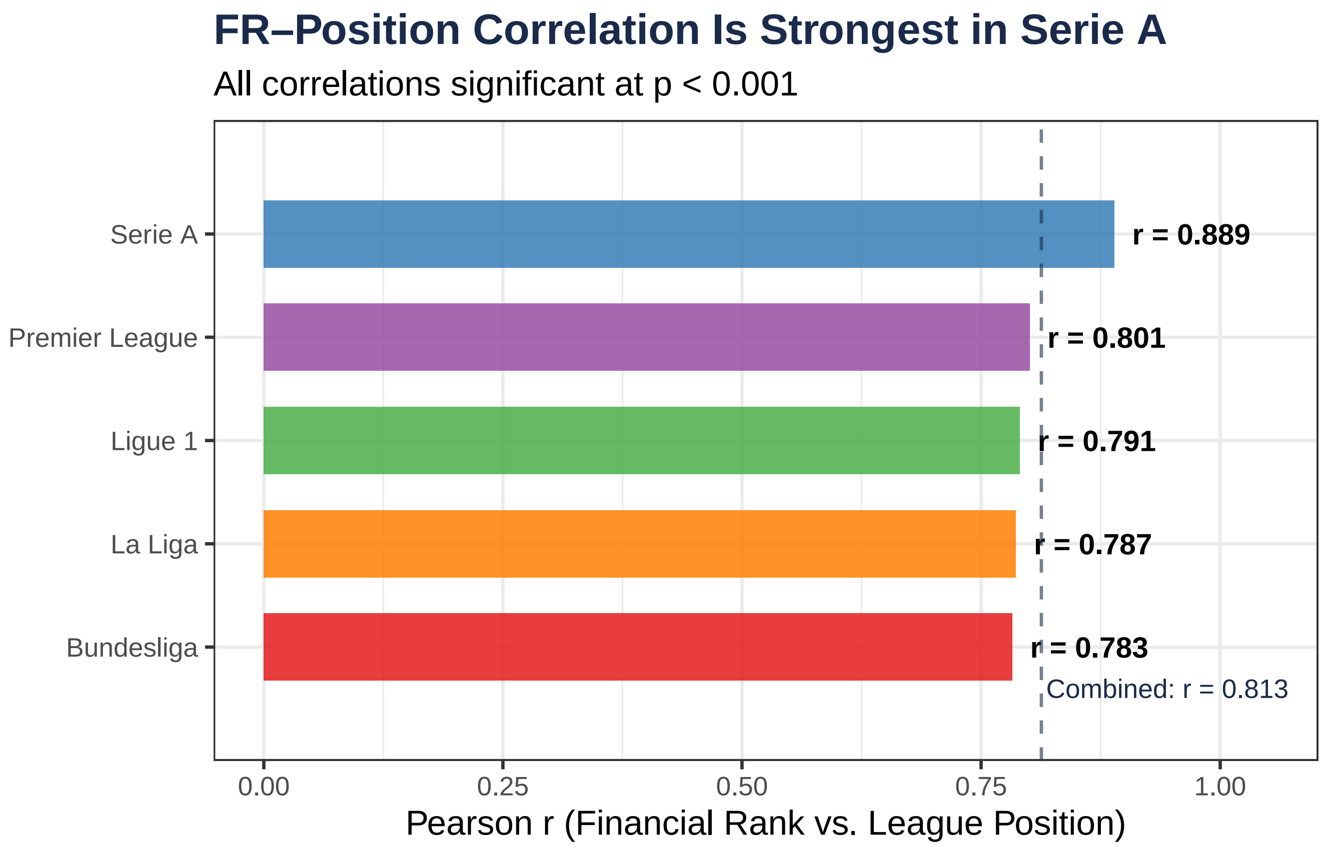

Figure 4: Figure 4. Pearson correlation between Financial Rank and League Position, by league. All are significant at $p < 0.001$.

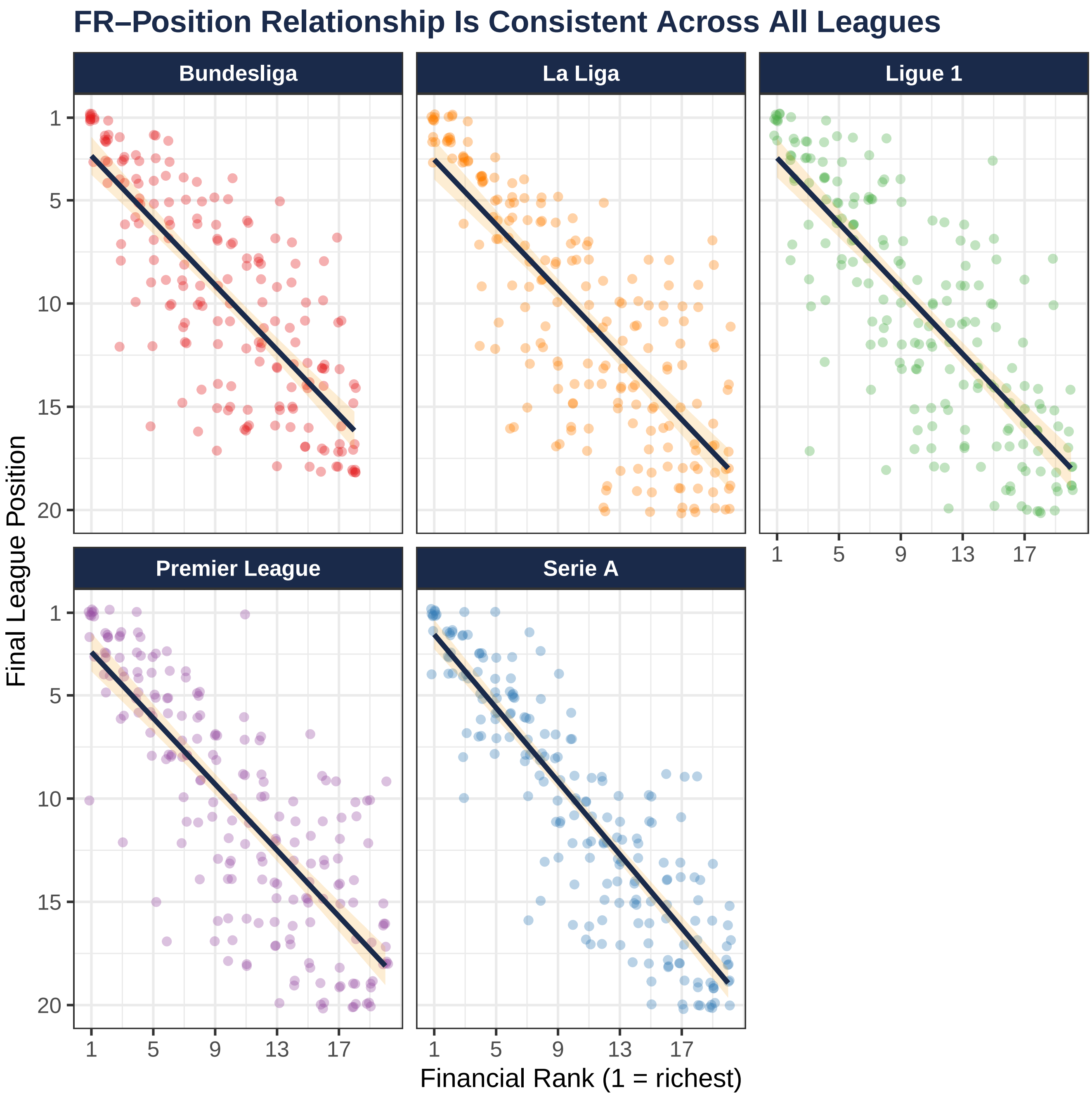

Figure 5: Figure 5. FR vs. Position faceted by league. Linear fits shown per league. The relationship is consistent in slope and significance across all five.

Table 4: Table 4. Model comparison.   = preferred specification (lowest AIC among OLS models).

Model	Formula	R^2	Adj. R^2	AIC	BIC
A	FR	0.6602	0.6599	5099.8	5114
B	FR + League FE	0.6604	0.6586	5107.4	5142
C	Log(MV)	0.5326	0.5322	5410.6	5425
D	Log(MV) + League + Season	0.6407	0.6385	5164.3	5203
G	FR × League (interaction)	0.6620	0.6588	5110.7	5164
H	**FR + FR²**	**0.6709**	**0.6703**	**5070.5**	**5090**
E	Champion ~ FR (Logit)	0.5477		182.4	

Note:

OLS response: League_Position (1 = champion). R^2 for Model E is McFadden pseudo-R^2.

Model H wins. Adding the quadratic term FR^2 reduces AIC from 5099.8 (Model A) to **5070.5**, a reduction of 29.3 points. The quadratic is confirmed by a nested F-test: $F(1, 972) = 31.71$, $p < 0.001$. Adding League fixed effects (Model B) or interaction terms (Model G) provides negligible improvement over the simple quadratic — the FR effect is **league-invariant**.

5.3 Model H: Estimated Coefficients

```r
# Read pre-computed coefficients (matches the fresh-fitted modH exactly)
coef_H_raw <- read_csv("outputs/20_quadratic_model_results.csv",
                       show_col_types = FALSE)

coef_H_raw |>
  mutate(
    term = recode(term,
      "(Intercept)"          = "Intercept ( )",
      "Financial_Rank"       = "Financial Rank ( )",
      "I(Financial_Rank^2)"  = "Financial Rank² ( )"
    ),
    across(c(estimate, std.error, statistic, conf.low, conf.high),
           ~ round(., 4)),
    p.value   = formatC(p.value, format = "e", digits = 2),
    CI_95     = sprintf("[%.4f, %.4f]", conf.low, conf.high)
  ) |>
  select(term, estimate, std.error, statistic, p.value, CI_95) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Term", "Estimate", "Std. Error", "t-value",
                  "p-value", "95% CI"),
    caption   = "Table 5. Model H coefficient estimates. Response: League Position (1 = champion)."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  column_spec(4:5, bold = TRUE)
```

Table 5: Table 5. Model H coefficient estimates. Response: League Position (1 = champion).

Term	Estimate	Std. Error	t-value	p-value	95% CI
Intercept ( )	0.4189	0.3442	**1.217**	**2.24e-01**	[-0.2557, 1.0936]
Financial Rank ( )	1.2305	0.0765	**16.082**	**8.83e-52**	[1.0806, 1.3805]
Financial Rank^2 ( )	-0.0202	0.0036	**-5.631**	**2.34e-08**	[-0.0273, -0.0132]

Interpreting the coefficients:

- **= 1.231:** Each one-rank increase in Financial Rank is associated with, on average, a **1.23-position worse finish**, holding the quadratic term constant. At FR = 1, the marginal effect of moving from 1st to 2nd richest is approximately $1.23 - 2 \times 0.020 \times 1 =$ **1.19 positions** worse.
- **$= -0.020$:** The negative quadratic coefficient indicates **diminishing returns at the bottom** — the positional penalty for each additional rank step shrinks as FR increases. At FR = 15, the marginal effect is $1.23 - 2 \times 0.020 \times 15 =$ **0.63 positions**.
- **R^2 = 0.671:** Financial Rank (linear + quadratic) explains **67.1%** of the variation in final league position.

5.4 Quadratic Fit Curve

```
pred_curve <- data.frame(Financial_Rank = fr_seq) |>
  bind_cols(
    predict(modH,
            newdata  = data.frame(Financial_Rank = fr_seq),
            interval = "confidence", level = 0.95) |>
      as.data.frame()
  )

ggplot(pred_curve, aes(x = Financial_Rank, y = fit)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), fill = "#2C3E73", alpha = 0.18) +
  geom_line(color = "#1A2A4A", linewidth = 1.7) +
  geom_point(
    data = data.frame(
      Financial_Rank = c(1, 5, 10, 15, 20),
      fit = predict(modH,
                    newdata = data.frame(Financial_Rank = c(1, 5, 10, 15, 20)))
    ),
    color = "#F5A623", size = 4, shape = 18
  ) +
  geom_text(
    data = data.frame(
      Financial_Rank = c(1, 5, 10, 15, 20),
      fit = predict(modH,
                    newdata = data.frame(Financial_Rank = c(1, 5, 10, 15, 20))),
      lbl = sprintf("FR=%d\n-> %s", c(1, 5, 10, 15, 20),
                    round(predict(modH,
                                  newdata = data.frame(
                                    Financial_Rank = c(1, 5, 10, 15, 20)
                                  )), 1))
    ),
```

```r
    aes(label = lbl), hjust = -0.15, size = 3.2, color = "#1A2A4A"
  ) +
  scale_y_reverse(
    breaks = seq(2, 20, 2),
    labels = c("2nd", "4th", "6th", "8th", "10th",
               "12th", "14th", "16th", "18th", "20th")
  ) +
  scale_x_continuous(breaks = seq(1, 20, 2),
                     labels = paste0("FR ", seq(1, 20, 2))) +
  labs(
    x       = "Financial Rank (1 = richest)",
    y       = "Predicted Final Position",
    title   = "Model H: Diminishing Returns at the Bottom of the Table",
    subtitle= sprintf(
      "Predicted position = %.3f + %.3f×FR − %.3f×FR² | R² = %.3f | AIC = %.1f",
      coef(modH)[1], coef(modH)[2], abs(coef(modH)[3]), r2_H, aic_H
    )
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", color = "#1A2A4A"),
    plot.subtitle = element_text(size = 10, color = "#566573"),
    panel.grid.minor = element_blank()
  )
```

5.5 League-Level Explanatory Power

```r
league_corr |>
  select(League, R2_FR, R2_LogMV) |>
  pivot_longer(cols = c(R2_FR, R2_LogMV),
               names_to = "Predictor", values_to = "R2") |>
  mutate(
    Predictor = recode(Predictor,
      "R2_FR"    = "Financial Rank (FR)",
      "R2_LogMV" = "Log(Market Value)"
    ),
    League = fct_reorder(League, R2, max)
  ) |>
  ggplot(aes(x = League, y = R2, fill = Predictor)) +
  geom_col(position = position_dodge(0.75), width = 0.65, alpha = 0.87) +
  geom_text(
    aes(label = sprintf("%.0f%%", R2 * 100)),
    position = position_dodge(0.75),
    vjust = -0.4, size = 3.5, fontface = "bold"
  ) +
  scale_fill_manual(values = c(
    "Financial Rank (FR)" = "#2C3E73",
    "Log(Market Value)"   = "#196F3D"
  )) +
  scale_y_continuous(labels = percent_format(accuracy = 1), limits = c(0, 1)) +
  labs(
```

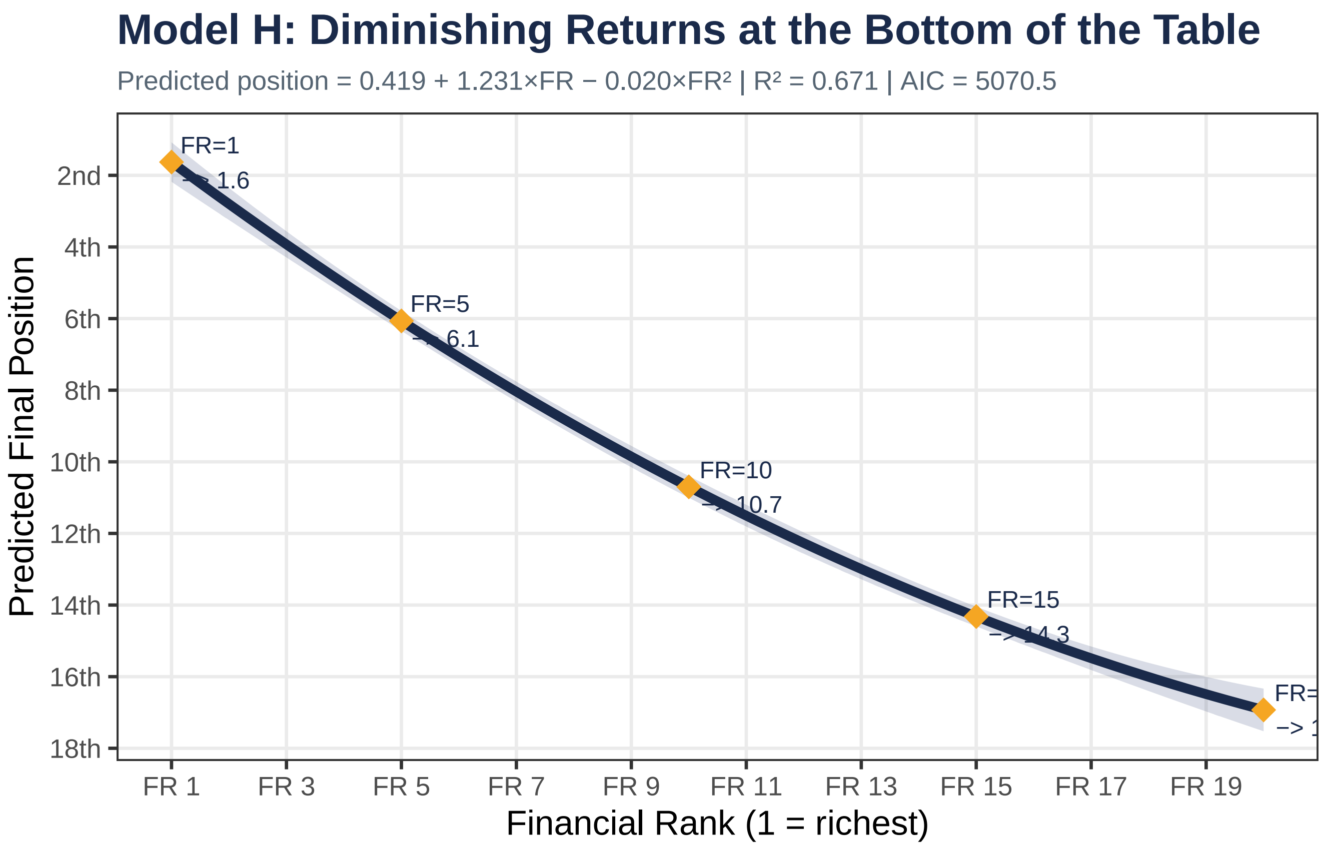

Figure 6: Figure 6. Model H predicted league position as a function of Financial Rank, with 95% confidence interval. y-axis reversed: lower = better.

```r
    x        = NULL,
    y        = "R² (variance explained)",
    fill     = "Predictor",
    title    = "Money Explains 61-79% of Final League Position",
    subtitle= "Financial Rank consistently outperforms Log(Market Value)"
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title     = element_text(face = "bold", color = "#1A2A4A"),
    legend.position  = "bottom",
    panel.grid.minor = element_blank()
  )
```

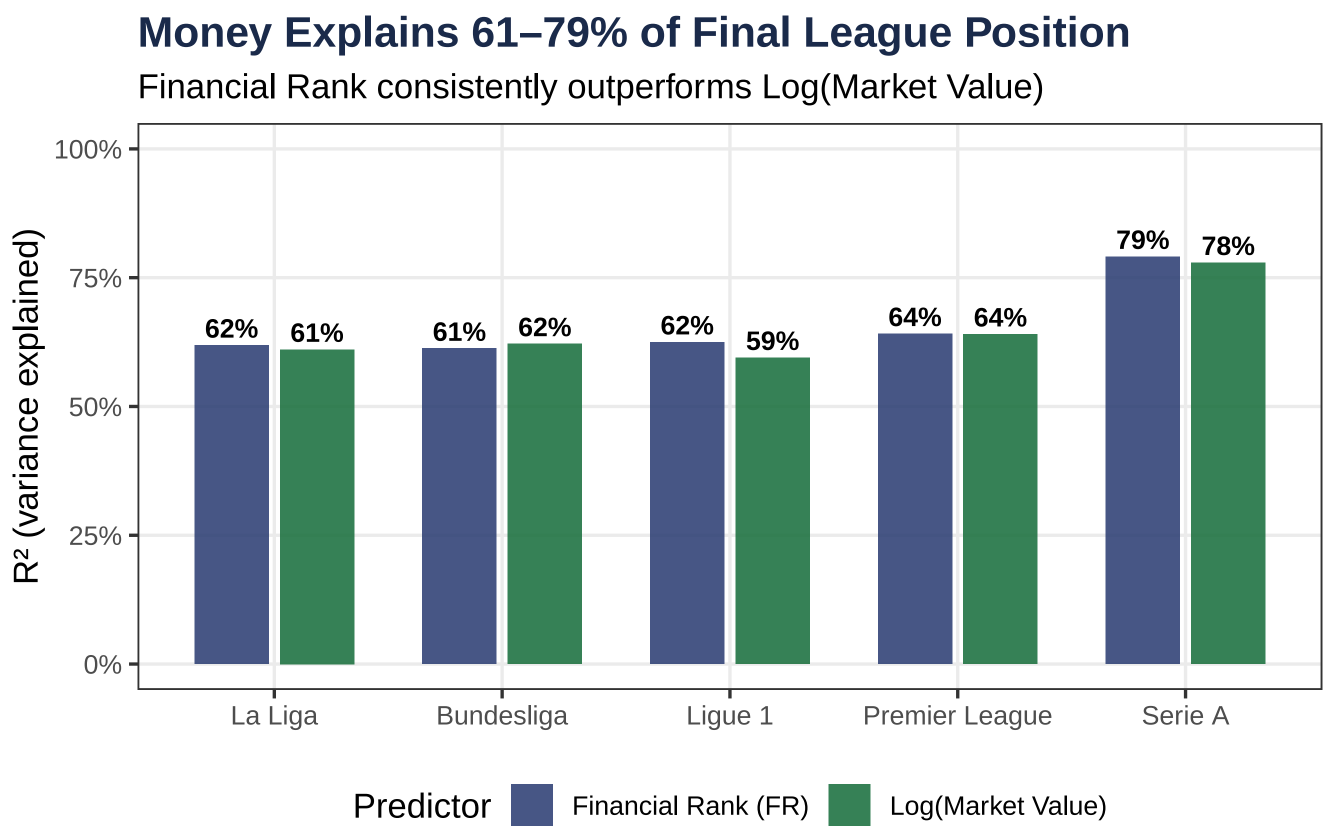

Figure 7: Figure 7. R^2 for Financial Rank predictor by league. Financial Rank alone explains 61–79% of final standings variation.

---

6 Champion Probability

6.1 Logistic Regression Model

Model E estimates the probability that a team wins the league title as a function of its Financial Rank using binary logistic regression.

$$\log\left(\frac{P(\text{Champion})}{1 - P(\text{Champion})}\right) = \alpha + \beta_{\text{FR}} \cdot \text{FR}$$

```
logit_coefs <- tidy(modE, conf.int = TRUE, exponentiate = FALSE) |>
  mutate(
    OR    = round(exp(estimate), 4),
    across(c(estimate, std.error, statistic, conf.low, conf.high), ~ round(., 4)),
    p.value = formatC(p.value, format = "e", digits = 2),
    term    = recode(term,
      "(Intercept)"    = "Intercept ( )",
      "Financial_Rank" = "Financial Rank ( )"
    )
  )

mcf_null   <- glm(Champion_bin ~ 1, data = df, family = binomial())
mcf_r2     <- round(1 - as.numeric(logLik(modE)) / as.numeric(logLik(mcf_null)), 3)

logit_coefs |>
  select(term, estimate, std.error, statistic, p.value, OR) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Term", "Log-Odds", "Std. Error", "z-value",
                  "p-value", "Odds Ratio"),
    caption = sprintf(
      "Table 6. Logistic Model E coefficients. McFadden R² = %.3f, AIC = %.1f.",
      mcf_r2, AIC(modE)
    )
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white")
```

Table 6: Table 6. Logistic Model E coefficients. McFadden R^2 = 0.548, AIC = 182.4.

Term	Log-Odds	Std. Error	z-value	p-value	Odds Ratio
Intercept ( )	1.642	0.4033	4.070	4.70e-05	5.1636
Financial Rank ( )	-1.298	0.1900	-6.834	8.24e-12	0.2729

Interpretation of = -1.298: Each one-step increase in Financial Rank multiplies the *odds* of winning the title by e^(-1.298) = **0.273** — a **72.7% reduction in the championship odds** per rank step.

6.2 Champion Probability Curve

```r
fr_range <- seq(1, max(df$Financial_Rank, na.rm = TRUE), length.out = 300)
prob_df  <- data.frame(Financial_Rank = fr_range) |>
  mutate(Prob = predict(modE, newdata = data.frame(Financial_Rank = fr_range),
                        type = "response"))

key_probs <- data.frame(Financial_Rank = c(1, 2, 3, 5, 10, 11)) |>
  mutate(Prob = predict(modE, newdata = data.frame(
    Financial_Rank = c(1, 2, 3, 5, 10, 11)
  ), type = "response"))

ggplot(prob_df, aes(x = Financial_Rank, y = Prob)) +
  geom_area(fill = "#C0392B", alpha = 0.12) +
  geom_line(color = "#C0392B", linewidth = 1.6) +
  geom_point(data = key_probs, aes(x = Financial_Rank, y = Prob),
             color = "#1A2A4A", size = 3.5, shape = 21, fill = "#F5A623") +
  geom_text(
    data = key_probs,
    aes(label = sprintf("FR=%d\n%.1f%%", Financial_Rank, Prob * 100)),
    vjust = -0.6, size = 3.2, color = "#1A2A4A", fontface = "bold"
  ) +
  scale_y_continuous(labels = percent_format(accuracy = 1),
                     breaks = seq(0, 0.7, 0.1)) +
  scale_x_continuous(breaks = 1:20) +
  labs(
    x       = "Financial Rank (1 = richest)",
    y       = "P(Winning the Title)",
    title   = sprintf("P(Champion | FR=1) = %.1f%% vs P(Champion | FR=11) < 0.01%%",
                      prob_fr1),
    subtitle= sprintf("Model E:   = %.3f (SE %.3f), p < 0.001, McFadden R² = %.3f",
                      coef(modE)[2], summary(modE)$coef[2,2], mcf_r2)
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", color = "#1A2A4A"),
    plot.subtitle = element_text(color = "#566573", size = 10),
    panel.grid.minor = element_blank()
  )
```

```r
data.frame(Financial_Rank = c(1, 2, 3, 4, 5, 8, 11)) |>
  mutate(
    Probability = predict(modE,
                          newdata = data.frame(Financial_Rank = c(1,2,3,4,5,8,11)),
                          type = "response"),
    Probability_pct = paste0(round(Probability * 100, 2), "%")
  ) |>
  select(Financial_Rank, Probability_pct) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Financial Rank", "P(Champion)"),
    caption   = "Table 7. Predicted championship probabilities at key FR values (Model E)."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
```

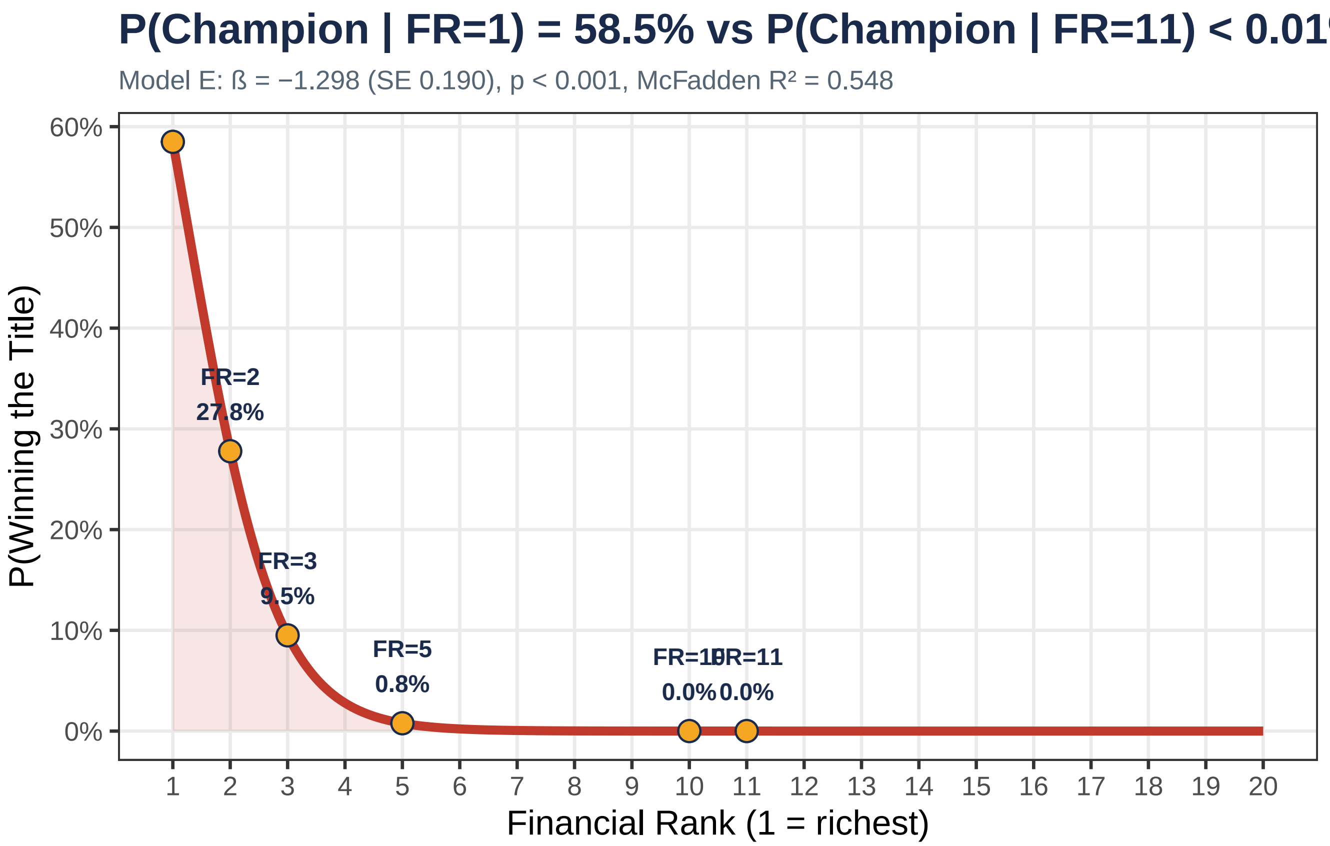

Figure 8: Figure 8. Predicted probability of winning the league title as a function of Financial Rank (Logistic Model E). The y-axis is probability (0 to 100%).

```r
  row_spec(1, bold = TRUE, background = "#FEF9E7")
```

Table 7: Table 7. Predicted championship probabilities at key FR values (Model E).

Financial Rank	P(Champion)
1	**58.5%**
2	27.78%
3	9.5%
4	2.79%
5	0.78%
8	0.02%
11	0%

FR=1 is not a guarantee: Even the richest squad has only a ~58.5% championship probability, reflecting the inherent randomness in football outcomes (injuries, fixture congestion, etc.). The model captures the *structural* advantage of financial strength, not individual season determinism.

---

7 Residual Analysis & Diagnostics

7.1 Residuals vs. Fitted Values

```r
# Identify story clubs for labelling
label_clubs <- aug_H |>
  filter(
    (Outlier_Flag != "Normal") |
    (grepl("Leicester|Leverkusen|Napoli|Lille|Liverpool", Team) &
     Champion_bin == 1)
  ) |>
  arrange(Residual_H)
```

```r
ggplot(aug_H, aes(x = Fitted_H, y = Residual_H)) +
  geom_hline(yintercept = 0, linetype = "solid",  color = "grey60", linewidth = 0.8) +
  geom_hline(yintercept = -1.5 * resid_sd, linetype = "dashed",
             color = "#1A2A4A", alpha = 0.7) +
  geom_hline(yintercept =  1.5 * resid_sd, linetype = "dashed",
             color = "#C0392B", alpha = 0.7) +
  geom_point(
    aes(color = Outlier_Flag), alpha = 0.55, size = 2
  ) +
  ggrepel::geom_text_repel(
    data  = label_clubs,
    aes(label = paste0(Team, "\n(", substr(Season, 3, 7), ")")),
    size  = 3, max.overlaps = 20, seed = 42,
    box.padding = 0.5, color = "#1A2A4A"
  ) +
```

```r
  annotate("text", x = 2, y = -1.5 * resid_sd - 0.5,
           label = sprintf("Overperformer threshold: -%.2f", 1.5 * resid_sd),
           hjust = 0, size = 3.2, color = "#1A2A4A") +
  annotate("text", x = 2, y =  1.5 * resid_sd + 0.5,
           label = sprintf("Underperformer threshold: +%.2f", 1.5 * resid_sd),
           hjust = 0, size = 3.2, color = "#C0392B") +
  scale_color_manual(
    values = c("Overperformer" = "#1A2A4A",
               "Normal"        = "grey70",
               "Underperformer"= "#C0392B"),
    labels = c("Overperformer"  = sprintf("Overperformer (n=%d)", n_overp),
               "Normal"         = sprintf("Normal (n=%d)",
                                          n_obs - n_overp - n_underp),
               "Underperformer" = sprintf("Underperformer (n=%d)", n_underp))
  ) +
  labs(
    x       = "Fitted Values (Predicted Position)",
    y       = "Residual (Actual - Predicted)",
    color   = NULL,
    title   = "Model H Residuals: Most Clubs Are Well-Predicted",
    subtitle= sprintf(
      "SD(residuals) = %.2f  |  ±1.5 SD threshold  |  N = %d",
      resid_sd, n_obs
    )
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", color = "#1A2A4A"),
    legend.position  = "bottom",
    panel.grid.minor = element_blank()
  )
```

> **Reading the residuals:** A *negative* residual means the team finished *better* than predicted by their financial rank. A *positive* residual means they underperformed relative to their resources. Leicester City's residual of **-10.5** — a team predicted to finish 11th, winning the league — is by far the most extreme in the dataset.

7.2 Cook's Distance

Cook's Distance measures the influence of each observation on the regression coefficients. Values above the threshold $4/N = 0.0041$ warrant closer inspection.

```r
high_cooks <- aug_H |>
  mutate(idx = row_number()) |>
  filter(CooksD > 4 / n_obs)

ggplot(aug_H |> mutate(idx = row_number()),
       aes(x = idx, y = CooksD)) +
  geom_segment(aes(xend = idx, yend = 0, color = CooksD > 4/n_obs),
               alpha = 0.6) +
  geom_point(aes(color = CooksD > 4 / n_obs), size = 1.8) +
  geom_hline(yintercept = 4 / n_obs, linetype = "dashed",
```

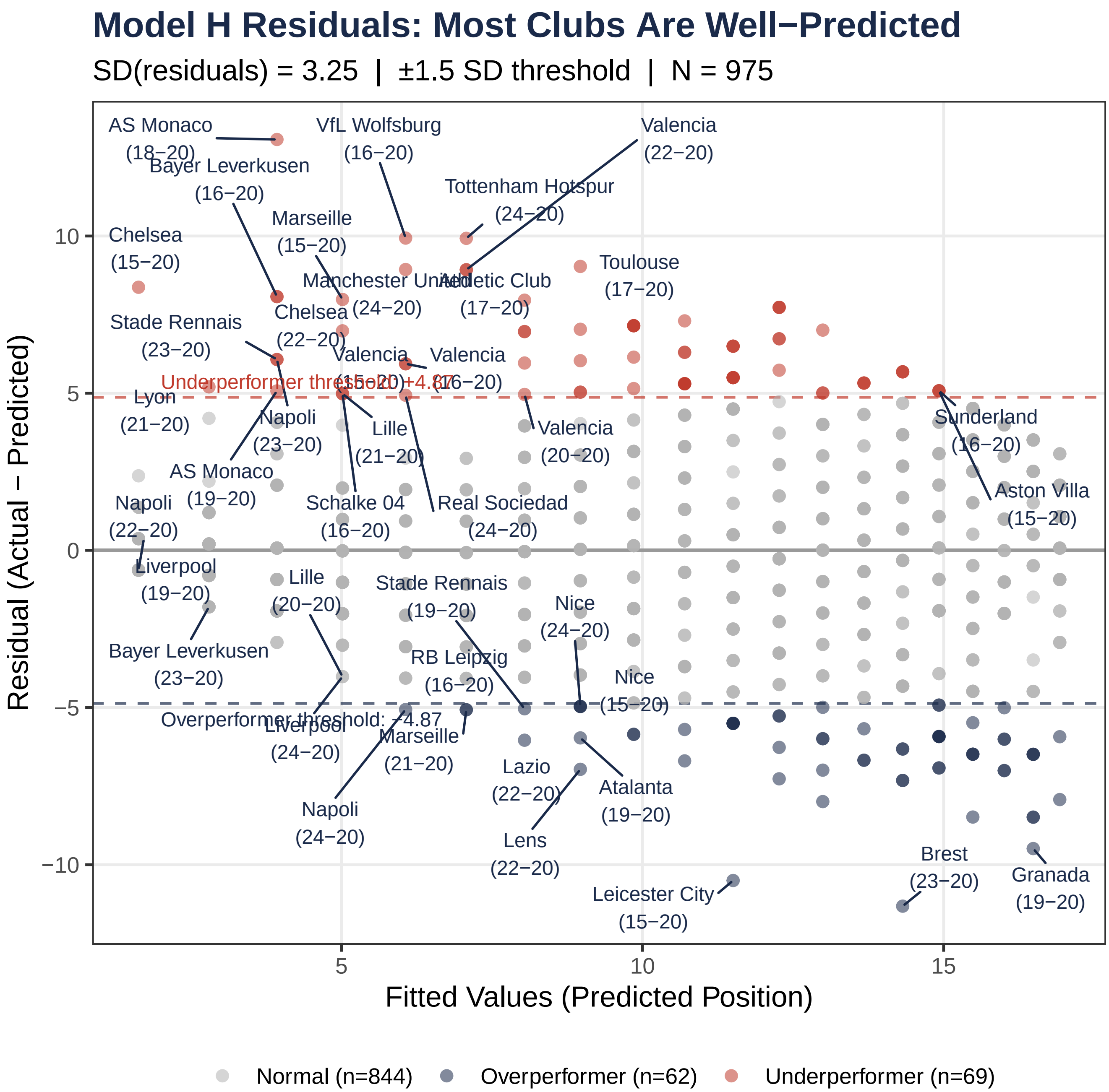

Figure 9: Figure 9. Residuals vs. Fitted Values for Model H. Dashed lines = ±1.5 SD threshold. Negative residual means the team finished better than predicted.

```r
             color = "#C0392B", linewidth = 1) +
  ggrepel::geom_text_repel(
    data  = high_cooks |> arrange(desc(CooksD)) |> head(12),
    aes(x = idx,
        label = paste0(Team, "\n", substr(Season, 3, 7))),
    size  = 2.8, max.overlaps = 15, seed = 99, color = "#1A2A4A",
    box.padding = 0.4
  ) +
  scale_color_manual(
    values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"),
    labels = c("FALSE" = "Below threshold",
               "TRUE"  = sprintf("Above 4/N = %.4f", 4/n_obs))
  ) +
  labs(
    x       = "Observation Index",
    y       = "Cook's Distance",
    color   = NULL,
    title   = "Cook's Distance Identifies Influential Observations",
    subtitle= sprintf(
      "Threshold = 4/N = %.4f  |  %d observations above threshold",
      cooks_thresh, n_influential
    )
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", color = "#1A2A4A"),
    legend.position  = "bottom",
    panel.grid.minor = element_blank()
  )
```

```r
aug_H |>
  filter(CooksD > 4 / n_obs) |>
  arrange(desc(CooksD)) |>
  select(Team, League, Season, League_Position, Financial_Rank,
         Fitted_H, Residual_H, CooksD) |>
  mutate(across(c(Fitted_H, Residual_H, CooksD), ~ round(., 3))) |>
  head(15) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Team", "League", "Season", "Actual Pos.",
                  "FR", "Predicted Pos.", "Residual", "Cook's D"),
    caption   = sprintf("Table 8. Top influential observations (Cook's D > %.4f).",
                        cooks_thresh)
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(1, bold = TRUE, background = "#FDEDEC")
```

Table 8: Table 8. Top influential observations (Cook's D > 0.0041).

Team	League	Season	Actual Pos.	FR	Predicted Pos.	Residual	Cook's I
AS Monaco	**Ligue 1**	**2018-2019**	**17**	**3**	**3.928**	**13.072**	**0.01**

Table 8: Table 8. Top influential observations (Cook's D > 0.0041).
(continued)

Team	League	Season	Actual Pos.	FR	Predicted Pos.	Residual	Cook's I
Sheffield United	Premier League	2019-2020	9	20	16.929	-7.929	0.01
Chelsea	Premier League	2015-2016	10	1	1.629	8.371	0.01
Granada	La Liga	2019-2020	7	19	16.489	-9.489	0.01
Rayo Vallecano	La Liga	2024-2025	8	19	16.489	-8.489	0.01
Clermont Foot	Ligue 1	2022-2023	8	19	16.489	-8.489	0.01
Las Palmas	La Liga	2015-2016	11	20	16.929	-5.929	0.01
Leicester City	Premier League	2015-2016	1	11	11.504	-10.504	0.00
Bastia	Ligue 1	2015-2016	10	19	16.489	-6.489	0.00
West Bromwich Albion	Premier League	2016-2017	10	19	16.489	-6.489	0.00
Fulham	Premier League	2022-2023	10	19	16.489	-6.489	0.00
Brest	Ligue 1	2023-2024	3	15	14.320	-11.320	0.00
Bayer Leverkusen	Bundesliga	2016-2017	12	3	3.928	8.072	0.00
Chelsea	Premier League	2022-2023	12	3	3.928	8.072	0.00
1. FC Union Berlin	Bundesliga	2020-2021	7	17	15.485	-8.485	0.00

7.3 Outlier Summary

```
aug_H |>
  filter(Outlier_Flag != "Normal") |>
  arrange(Residual_H) |>
  select(Team, League, Season, League_Position, Financial_Rank,
         Fitted_H, Residual_H, Outlier_Flag) |>
  mutate(across(c(Fitted_H, Residual_H), ~ round(., 2))) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Team", "League", "Season", "Actual",
                  "FR", "Predicted", "Residual", "Category"),
    caption   = sprintf(
      "Table 9. Statistical outliers (|Residual| > 1.5 SD = %.2f): %d overperformers, %d underperformer
      1.5 * resid_sd, n_overp, n_underp
    )
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(which(
    aug_H |> filter(Outlier_Flag != "Normal") |>
      arrange(Residual_H) |> pull(Outlier_Flag) == "Overperformer"
  ), background = "#EBF5FB") |>
  row_spec(which(
    aug_H |> filter(Outlier_Flag != "Normal") |>
      arrange(Residual_H) |> pull(Outlier_Flag) == "Underperformer"
  ), background = "#FDEDEC")
```

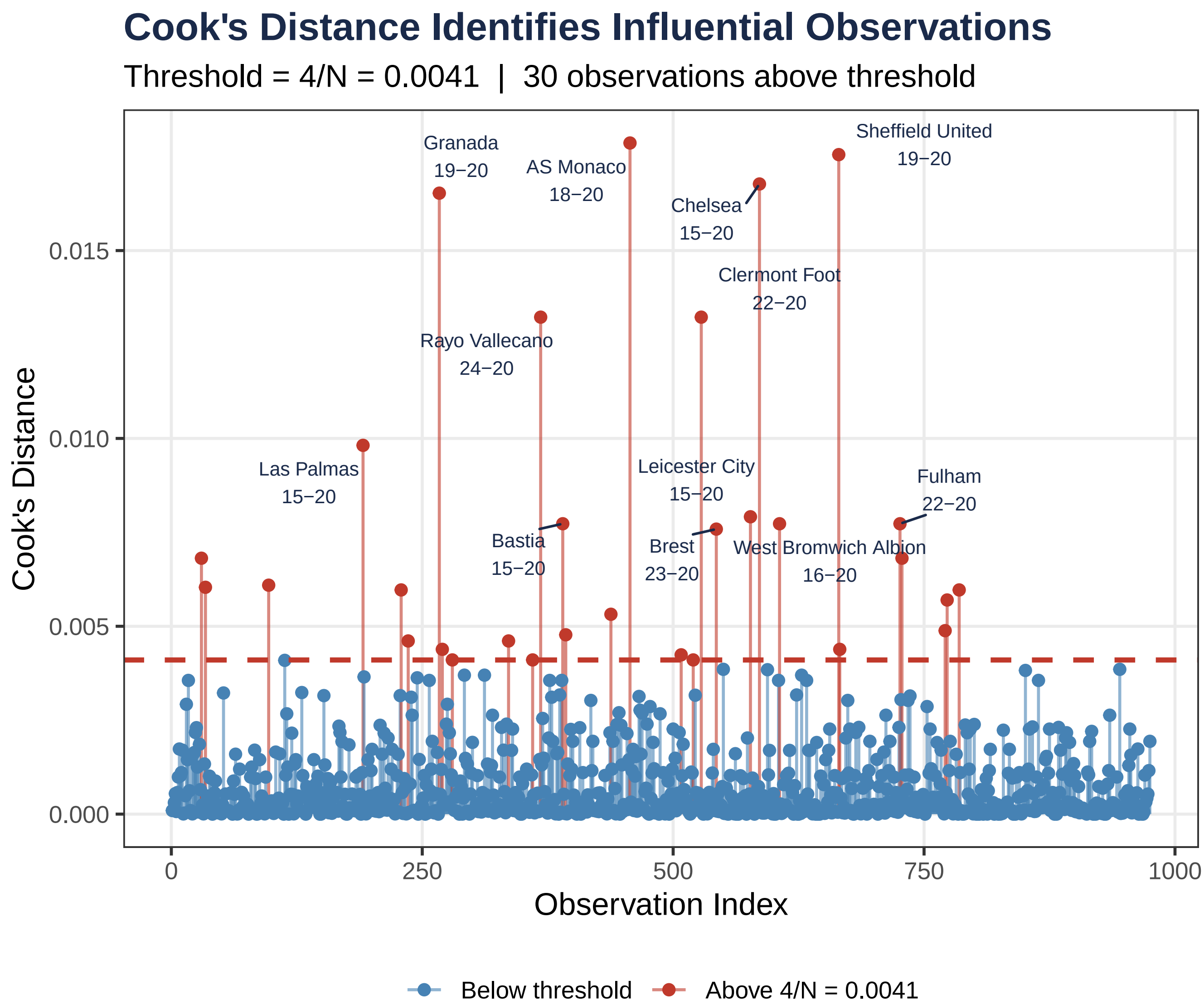

Figure 10: Figure 10. Cook's Distance for all 975 observations. The dashed red line is the threshold 4/N = 0.0041. Labelled points exceed the threshold.

Table 9: Table 9. Statistical outliers (|Residual| > 1.5 SD = 4.87): 62 overperformers, 69 underperformers.

Team	League	Season	Actual	FR	Predicted	Residual	Category
Brest	Ligue 1	2023-2024	3	15	14.32	-11.32	Overperformer
Leicester City	Premier League	2015-2016	1	11	11.50	-10.50	Overperformer
Granada	La Liga	2019-2020	7	19	16.49	-9.49	Overperformer
Rayo Vallecano	La Liga	2024-2025	8	19	16.49	-8.49	Overperformer
Clermont Foot	Ligue 1	2022-2023	8	19	16.49	-8.49	Overperformer
1. FC Union Berlin	Bundesliga	2020-2021	7	17	15.49	-8.49	Overperformer
1. FC Union Berlin	Bundesliga	2021-2022	5	13	12.99	-7.99	Overperformer
Sheffield United	Premier League	2019-2020	9	20	16.93	-7.93	Overperformer
Caen	Ligue 1	2015-2016	7	15	14.32	-7.32	Overperformer
Burnley	Premier League	2017-2018	7	15	14.32	-7.32	Overperformer
Getafe	La Liga	2018-2019	5	12	12.27	-7.27	Overperformer
Eibar	La Liga	2017-2018	9	18	16.01	-7.01	Overperformer
Chievo Verona	Serie A	2015-2016	9	18	16.01	-7.01	Overperformer
Stade de Reims	Ligue 1	2019-2020	6	13	12.99	-6.99	Overperformer
Lens	Ligue 1	2022-2023	2	8	8.97	-6.97	Overperformer
1. FC Heidenheim 1846	Bundesliga	2023-2024	8	16	14.92	-6.92	Overperformer
Getafe	La Liga	2017-2018	8	16	14.92	-6.92	Overperformer
1. FC Union Berlin	Bundesliga	2022-2023	4	10	10.70	-6.70	Overperformer
FC Cologne	Bundesliga	2021-2022	7	14	13.68	-6.68	Overperformer
Lens	Ligue 1	2020-2021	7	14	13.68	-6.68	Overperformer
Bastia	Ligue 1	2015-2016	10	19	16.49	-6.49	Overperformer
West Bromwich Albion	Premier League	2016-2017	10	19	16.49	-6.49	Overperformer
Fulham	Premier League	2022-2023	10	19	16.49	-6.49	Overperformer
Angers	Ligue 1	2015-2016	9	17	15.49	-6.49	Overperformer
AFC Bournemouth	Premier League	2016-2017	9	17	15.49	-6.49	Overperformer
Hellas Verona	Serie A	2019-2020	9	17	15.49	-6.49	Overperformer
Espanyol	La Liga	2016-2017	8	15	14.32	-6.32	Overperformer
Stade de Reims	Ligue 1	2018-2019	8	15	14.32	-6.32	Overperformer
Montpellier	Ligue 1	2018-2019	6	12	12.27	-6.27	Overperformer
Lazio	Serie A	2022-2023	2	7	8.04	-6.04	Overperformer
Osasuna	La Liga	2019-2020	10	18	16.01	-6.01	Overperformer
Burnley	Premier League	2019-2020	10	18	16.01	-6.01	Overperformer
SC Freiburg	Bundesliga	2016-2017	7	13	12.99	-5.99	Overperformer
Nantes	Ligue 1	2016-2017	7	13	12.99	-5.99	Overperformer
Atalanta	Serie A	2019-2020	3	8	8.97	-5.97	Overperformer
Las Palmas	La Liga	2015-2016	11	20	16.93	-5.93	Overperformer
Mallorca	La Liga	2022-2023	9	16	14.92	-5.92	Overperformer
Leeds United	Premier League	2020-2021	9	16	14.92	-5.92	Overperformer
Brentford	Premier League	2022-2023	9	16	14.92	-5.92	Overperformer
Sampdoria	Serie A	2020-2021	9	16	14.92	-5.92	Overperformer
Saint-Étienne	Ligue 1	2018-2019	4	9	9.85	-5.85	Overperformer
Atalanta	Serie A	2016-2017	4	9	9.85	-5.85	Overperformer
SC Freiburg	Bundesliga	2024-2025	5	10	10.70	-5.70	Overperformer
Werder Bremen	Bundesliga	2024-2025	8	14	13.68	-5.68	Overperformer
Hertha Berlin	Bundesliga	2016-2017	6	11	11.50	-5.50	Overperformer
Mainz	Bundesliga	2024-2025	6	11	11.50	-5.50	Overperformer
Strasbourg	Ligue 1	2021-2022	6	11	11.50	-5.50	Overperformer
West Ham United	Premier League	2020-2021	6	11	11.50	-5.50	Overperformer

Table 9: Table 9. Statistical outliers (|Residual| > 1.5 SD = 4.87): 62 overperformers, 69 underperformers. *(continued)*

Team	League	Season	Actual	FR	Predicted	Residual	Category
Mallorca	La Liga	2024-2025	10	17	15.49	-5.49	Overperformer
Wolverhampton Wanderers	Premier League	2018-2019	7	12	12.27	-5.27	Overperformer
Nottingham Forest	Premier League	2024-2025	7	12	12.27	-5.27	Overperformer
RB Leipzig	Bundesliga	2016-2017	2	6	7.07	-5.07	Overperformer
Marseille	Ligue 1	2021-2022	2	6	7.07	-5.07	Overperformer
Napoli	Serie A	2024-2025	1	5	6.07	-5.07	Overperformer
Stade Rennais	Ligue 1	2019-2020	3	7	8.04	-5.04	Overperformer
Crystal Palace	Premier League	2022-2023	11	18	16.01	-5.01	Overperformer
Montpellier	Ligue 1	2019-2020	8	13	12.99	-4.99	Overperformer
TSG Hoffenheim	Bundesliga	2016-2017	4	8	8.97	-4.97	Overperformer
Nice	Ligue 1	2015-2016	4	8	8.97	-4.97	Overperformer
Nice	Ligue 1	2024-2025	4	8	8.97	-4.97	Overperformer
Fortuna Düsseldorf	Bundesliga	2018-2019	10	16	14.92	-4.92	Overperformer
Eibar	La Liga	2016-2017	10	16	14.92	-4.92	Overperformer
Real Sociedad	La Liga	2024-2025	11	5	6.07	4.93	Underperformer
Valencia	La Liga	2020-2021	13	7	8.04	4.96	Underperformer
Schalke 04	Bundesliga	2016-2017	10	4	5.02	4.98	Underperformer
Lille	Ligue 1	2021-2022	10	4	5.02	4.98	Underperformer
Schalke 04	Bundesliga	2020-2021	18	13	12.99	5.01	Underperformer
Leganés	La Liga	2019-2020	18	13	12.99	5.01	Underperformer
Hertha Berlin	Bundesliga	2020-2021	14	8	8.97	5.03	Underperformer
Aston Villa	Premier League	2021-2022	14	8	8.97	5.03	Underperformer
AS Monaco	Ligue 1	2019-2020	9	3	3.93	5.07	Underperformer
Aston Villa	Premier League	2015-2016	20	16	14.92	5.08	Underperformer
Sunderland	Premier League	2016-2017	20	16	14.92	5.08	Underperformer
West Bromwich Albion	Premier League	2017-2018	20	16	14.92	5.08	Underperformer
TSG Hoffenheim	Bundesliga	2024-2025	15	9	9.85	5.15	Underperformer
Lyon	Ligue 1	2021-2022	8	2	2.80	5.20	Underperformer
Deportivo La Coruña	La Liga	2016-2017	16	10	10.70	5.30	Underperformer
Girona	La Liga	2024-2025	16	10	10.70	5.30	Underperformer
Stade de Reims	Ligue 1	2024-2025	16	10	10.70	5.30	Underperformer
West Ham United	Premier League	2019-2020	16	10	10.70	5.30	Underperformer
Cagliari	Serie A	2020-2021	16	10	10.70	5.30	Underperformer
Espanyol	La Liga	2022-2023	19	14	13.68	5.32	Underperformer
Stoke City	Premier League	2017-2018	19	14	13.68	5.32	Underperformer
Fulham	Premier League	2018-2019	19	14	13.68	5.32	Underperformer
Celta Vigo	La Liga	2019-2020	17	11	11.50	5.50	Underperformer
Lille	Ligue 1	2017-2018	17	11	11.50	5.50	Underperformer
Udinese	Serie A	2015-2016	17	11	11.50	5.50	Underperformer
Torino	Serie A	2020-2021	17	11	11.50	5.50	Underperformer
Eibar	La Liga	2020-2021	20	15	14.32	5.68	Underperformer
Toulouse	Ligue 1	2019-2020	20	15	14.32	5.68	Underperformer
Parma	Serie A	2020-2021	20	15	14.32	5.68	Underperformer
Stade de Reims	Ligue 1	2015-2016	18	12	12.27	5.73	Underperformer
VfL Wolfsburg	Bundesliga	2021-2022	12	5	6.07	5.93	Underperformer
Valencia	La Liga	2016-2017	12	5	6.07	5.93	Underperformer
Bordeaux	Ligue 1	2018-2019	14	7	8.04	5.96	Underperformer
Sampdoria	Serie A	2015-2016	15	8	8.97	6.03	Underperformer

Table 9: Table 9. Statistical outliers (|Residual| > 1.5 SD = 4.87): 62 overperformers, 69 underperformers. *(continued)*

Team	League	Season	Actual	FR	Predicted	Residual	Category
Stade Rennais	Ligue 1	2023-2024	10	3	3.93	6.07	Underperformer
Napoli	Serie A	2023-2024	10	3	3.93	6.07	Underperformer
Everton	Premier League	2021-2022	16	9	9.85	6.15	Underperformer
Saint-Étienne	Ligue 1	2019-2020	17	10	10.70	6.30	Underperformer
Nottingham Forest	Premier League	2023-2024	17	10	10.70	6.30	Underperformer
Lorient	Ligue 1	2016-2017	18	11	11.50	6.50	Underperformer
AFC Bournemouth	Premier League	2019-2020	18	11	11.50	6.50	Underperformer
Leicester City	Premier League	2022-2023	18	11	11.50	6.50	Underperformer
Las Palmas	La Liga	2017-2018	19	12	12.27	6.73	Underperformer
Las Palmas	La Liga	2024-2025	19	12	12.27	6.73	Underperformer
TSG Hoffenheim	Bundesliga	2015-2016	15	7	8.04	6.96	Underperformer
Real Betis	La Liga	2019-2020	15	7	8.04	6.96	Underperformer
Valencia	La Liga	2015-2016	12	4	5.02	6.98	Underperformer
Southampton	Premier League	2022-2023	20	13	12.99	7.01	Underperformer
VfL Wolfsburg	Bundesliga	2017-2018	16	8	8.97	7.03	Underperformer
VfB Stuttgart	Bundesliga	2015-2016	17	9	9.85	7.15	Underperformer
Celta Vigo	La Liga	2018-2019	17	9	9.85	7.15	Underperformer
Sevilla	La Liga	2024-2025	17	9	9.85	7.15	Underperformer
Southampton	Premier League	2017-2018	17	9	9.85	7.15	Underperformer
Newcastle United	Premier League	2015-2016	18	10	10.70	7.30	Underperformer
Espanyol	La Liga	2019-2020	20	12	12.27	7.73	Underperformer
Granada	La Liga	2023-2024	20	12	12.27	7.73	Underperformer
Bordeaux	Ligue 1	2021-2022	20	12	12.27	7.73	Underperformer
Fiorentina	Serie A	2018-2019	16	7	8.04	7.96	Underperformer
Marseille	Ligue 1	2015-2016	13	4	5.02	7.98	Underperformer
Bayer Leverkusen	Bundesliga	2016-2017	12	3	3.93	8.07	Underperformer
Chelsea	Premier League	2022-2023	12	3	3.93	8.07	Underperformer
Chelsea	Premier League	2015-2016	10	1	1.63	8.37	Underperformer
Athletic Club	La Liga	2017-2018	16	6	7.07	8.93	Underperformer
Valencia	La Liga	2022-2023	16	6	7.07	8.93	Underperformer
Manchester United	Premier League	2024-2025	15	5	6.07	8.93	Underperformer
Toulouse	Ligue 1	2017-2018	18	8	8.97	9.03	Underperformer
Tottenham Hotspur	Premier League	2024-2025	17	6	7.07	9.93	Underperformer
VfL Wolfsburg	Bundesliga	2016-2017	16	5	6.07	9.93	Underperformer
AS Monaco	Ligue 1	2018-2019	17	3	3.93	13.07	Underperformer

---

8 The Exceptions

8.1 Champions from Outside the Financial Top Three

```
surprise_raw <- read_csv("outputs/07_champions_surprise_detail.csv",
                         show_col_types = FALSE)

surprise_aug <- aug_H |>
```

```r
  filter(Champion_bin == 1, Financial_Rank > 3) |>
  select(League, Season, Team, Financial_Rank, Market_Value_M_EUR,
         Fitted_H, Residual_H) |>
  mutate(across(c(Fitted_H, Residual_H), ~ round(., 1)))

surprise_aug |>
  arrange(Residual_H) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("League", "Season", "Club", "FR",
                  "Squad Value (M€)", "Model Prediction", "Residual"),
    caption   = sprintf(
      "Table 10. The %d champions who won from outside the financial top 3 (FR > 3).",
      fr_gt3_n
    )
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(which(surprise_aug |> arrange(Residual_H) |> pull(Team) == "Leicester City"),
           bold = TRUE, background = "#FDEDEC") |>
  column_spec(4, bold = TRUE, color = "#C0392B")
```

Table 10: Table 10. The 4 champions who won from outside the financial top 3 (FR > 3).

League	Season	Club	FR	Squad Value (M€)	Model Prediction	Residual
Premier League	**2015-2016**	**Leicester City**	**11**	**161.1**	**11.5**	**-10.5**
Serie A	2024-2025	Napoli	**5**	500.4	6.1	-5.1
Ligue 1	2020-2021	Lille	**4**	323.2	5.0	-4.0
Premier League	2024-2025	Liverpool	**4**	951.0	5.0	-4.0

Only **4 of 50 champions** (8%) came from outside the financial top 3. Of these, only **Leicester City** (FR = 11) exceeded the ± 1.5 SD statistical outlier threshold. The other three surprise champions (Napoli FR=5, Lille FR=4, Liverpool FR=4) are within one standard deviation of expectation — financially competitive clubs who happened to win.

8.2 All 50 Champions: Residual Dotplot

```r
champ_csv <- read_csv("outputs/31_all_champions_with_residuals.csv",
                      show_col_types = FALSE) |>
  mutate(
    Label    = paste0(Team, " (", substr(Season, 1, 7), ", ",
                      substr(League, 1, 3), ")"),
    Surprise = Financial_Rank > 3,
    LeagueF  = factor(as.character(League), levels = league_order)
  ) |>
  arrange(Residual_H)

ggplot(champ_csv, aes(x = Residual_H,
                      y = reorder(Label, -Residual_H))) +
  annotate("rect",
```

```
        xmin = -Inf, xmax = -1.5 * resid_sd,
        ymin = -Inf, ymax = Inf,
        fill = "#1A2A4A", alpha = 0.06) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey60") +
  geom_vline(xintercept = -1.5 * resid_sd, linetype = "dotted",
             color = "#1A2A4A", alpha = 0.6) +
  geom_segment(aes(x = 0, xend = Residual_H,
                   yend = reorder(Label, -Residual_H),
                   color = LeagueF), alpha = 0.6) +
  geom_point(aes(color = LeagueF,
                 shape = Surprise, size = Surprise)) +
  ggrepel::geom_text_repel(
    data = champ_csv |> filter(Surprise),
    aes(label = sprintf("FR=%d", Financial_Rank)),
    size = 3, nudge_x = -0.5, color = "#C0392B", fontface = "bold",
    direction = "y", seed = 7
  ) +
  scale_color_manual(values = league_colours) +
  scale_shape_manual(values = c("FALSE" = 16, "TRUE" = 18),
                     labels = c("FALSE" = "FR   3", "TRUE" = "FR > 3 (surprise)"),
                     name = "Champion type") +
  scale_size_manual(values = c("FALSE" = 2.5, "TRUE" = 5), guide = "none") +
  labs(
    x       = "Residual: Actual - Predicted Position (negative = beat the model)",
    y       = NULL,
    color   = "League",
    title   = "All 50 Champions Beat Their Financial Prediction",
    subtitle= sprintf(
      "Every champion has a negative residual. Blue zone: statistical outlier (< -%.2f)",
      1.5 * resid_sd
    )
  ) +
  theme_bw(base_size = 10) +
  theme(
    plot.title     = element_text(face = "bold", color = "#1A2A4A"),
    legend.position  = "bottom",
    axis.text.y    = element_text(size = 7),
    panel.grid.minor = element_blank()
  )
```

8.3 Champion FR Distribution

```
champ_csv |>
  mutate(
    FR_Group = case_when(
      Financial_Rank == 1        ~ "FR 1 (richest)",
      Financial_Rank == 2        ~ "FR 2",
      Financial_Rank == 3        ~ "FR 3",
      Financial_Rank %in% 4:5    ~ "FR 4-5",
      TRUE                        ~ "FR 6+"
    ),
```


All 50 Champions Beat Their Financial Prediction
Every champion has a negative residual. Blue zone: statistical outlier (< −4.87)
Leicester City (2015−20, Pre)
Napoli (2024−20, Ser)
Liverpool (2024−20, Pre)
Lille (2020−20, Lig)
Atlético Madrid (2020−20, La)
AC Milan (2021−20, Ser)
Real Madrid (2019−20, La)
Liverpool (2019−20, Pre)
Bayer Leverkusen (2023−20, Bun)
Barcelona (2024−20, La)
Barcelona (2022−20, La)
AS Monaco (2016−20, Lig)
Real Madrid (2023−20, La)
Real Madrid (2021−20, La)
Real Madrid (2016−20, La)
Paris Saint−Germain (2024−20, Lig)
Paris Saint−Germain (2023−20, Lig)
Paris Saint−Germain (2022−20, Lig)
Paris Saint−Germain (2021−20, Lig)
Paris Saint−Germain (2019−20, Lig)
Paris Saint−Germain (2018−20, Lig)
Paris Saint−Germain (2017−20, Lig)
Paris Saint−Germain (2015−20, Lig)
Napoli (2022−20, Ser)
Manchester City (2023−20, Pre)
Manchester City (2022−20, Pre)
Manchester City (2021−20, Pre)
Manchester City (2020−20, Pre)
Manchester City (2018−20, Pre)
Manchester City (2017−20, Pre)
Juventus (2019−20, Ser)
Juventus (2018−20, Ser)
Juventus (2017−20, Ser)
Juventus (2016−20, Ser)
Juventus (2015−20, Ser)
Internazionale (2023−20, Ser)
Internazionale (2020−20, Ser)
Chelsea (2016−20, Pre)
Bayern Munich (2024−20, Bun)
Bayern Munich (2022−20, Bun)
Bayern Munich (2021−20, Bun)
Bayern Munich (2020−20, Bun)
Bayern Munich (2019−20, Bun)
Bayern Munich (2018−20, Bun)
Bayern Munich (2017−20, Bun)
Bayern Munich (2016−20, Bun)
Bayern Munich (2015−20, Bun)
Barcelona (2018−20, La)
Barcelona (2017−20, La)
Barcelona (2015−20, La)
FR=11
FR=5
FR=4
FR=4
−9
−6
−3
0
Residual: Actual − Predicted Position (negative = beat the model)
Champion type
FR = 3
FR > 3 (surprise)
League
Bundesliga
La Liga
Ligue 1
Prem

```
    FR_Group = factor(FR_Group,
                      levels = c("FR 1 (richest)","FR 2","FR 3","FR 4-5","FR 6+"))
  ) |>
  count(FR_Group) |>
  mutate(
    Pct      = n / sum(n),
    Surprise = FR_Group %in% c("FR 4-5", "FR 6+")
  ) |>
  ggplot(aes(x = FR_Group, y = n, fill = Surprise)) +
  geom_col(width = 0.65, alpha = 0.87) +
  geom_text(aes(label = sprintf("%d\n(%.0f%%)", n, Pct * 100)),
            vjust = -0.3, fontface = "bold", size = 4) +
  scale_fill_manual(values = c("FALSE" = "#2C3E73", "TRUE" = "#C0392B"),
                    labels = c("FALSE" = "Expected zone (FR   3)",
                               "TRUE"  = "Surprise zone (FR > 3)")) +
  scale_y_continuous(limits = c(0, 45)) +
  labs(
    x        = "Financial Rank Group",
    y        = "Number of Titles",
    fill     = NULL,
    title    = sprintf("%d%% of Titles Were Won by the League's Richest Club",
                      pct_fr1),
    subtitle= "10 seasons × 5 leagues = 50 champions total"
  ) +
  theme_bw(base_size = 13) +
  theme(
    plot.title      = element_text(face = "bold", color = "#1A2A4A"),
    legend.position  = "bottom",
    panel.grid.minor = element_blank()
  )
```

8.4 Leicester City 2015–16: A Closer Look

```
pl_data <- aug_H |>
  filter(as.character(League) == "Premier League") |>
  mutate(
    is_leicester = grepl("Leicester", Team) & grepl("2015", Season),
    pt_label     = ifelse(is_leicester,
                          "Leicester City\n2015-16\n(FR=11, Pos=1)",
                          NA_character_)
  )

ggplot(pl_data, aes(x = Financial_Rank, y = League_Position)) +
  geom_jitter(aes(color = is_leicester, alpha = is_leicester, size = is_leicester),
              width = 0.15, height = 0.15) +
  geom_smooth(data = pl_data |> filter(!is_leicester),
              method = "lm", formula = y ~ x + I(x^2),
              color = "#1A2A4A", fill = "#1A2A4A", alpha = 0.1,
              se = TRUE, linewidth = 1.2) +
  ggrepel::geom_text_repel(
    data = pl_data |> filter(is_leicester),
```

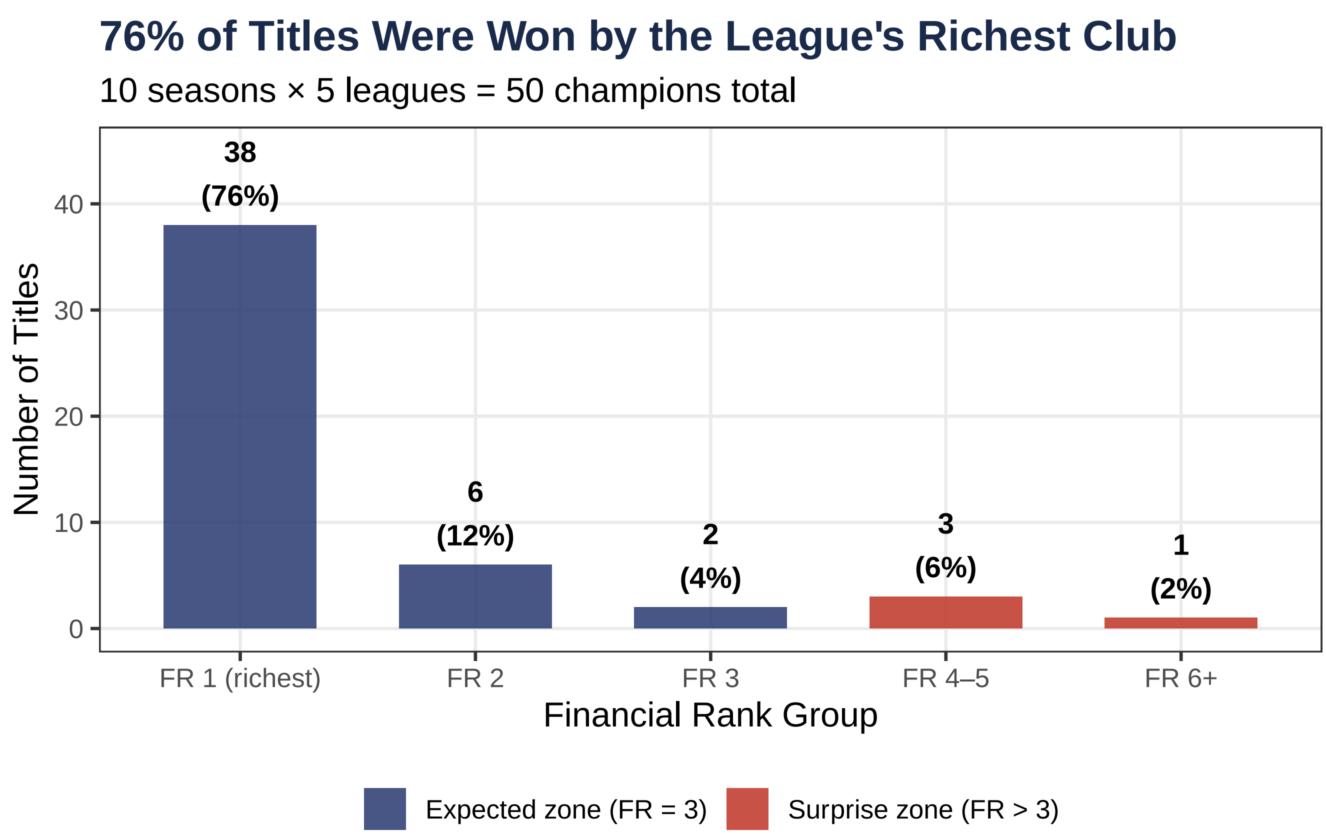

Figure 12: Figure 12. Distribution of Financial Rank among all 50 league champions.

```r
    aes(label = pt_label),
    size = 4, fontface = "bold", color = "#C0392B",
    nudge_x = -2, nudge_y = -2, box.padding = 0.8, seed = 123
  ) +
  scale_y_reverse(breaks = c(1, 5, 10, 15, 20)) +
  scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"),
                     guide = "none") +
  scale_alpha_manual(values = c("FALSE" = 0.4, "TRUE" = 1), guide = "none") +
  scale_size_manual(values = c("FALSE" = 2, "TRUE" = 6), guide = "none") +
  labs(
    x        = "Financial Rank",
    y        = "Final League Position",
    title    = "Leicester City 2015-16: A Premier League Outlier",
    subtitle= sprintf("Red = Leicester (FR=11, actual=1, predicted=%.1f, residual=%.1f)",
                      leicester_row$Fitted_H[1], leicester_row$Residual_H[1])
  ) +
  theme_bw(base_size = 13) +
  theme(plot.title = element_text(face = "bold", color = "#1A2A4A"))
```

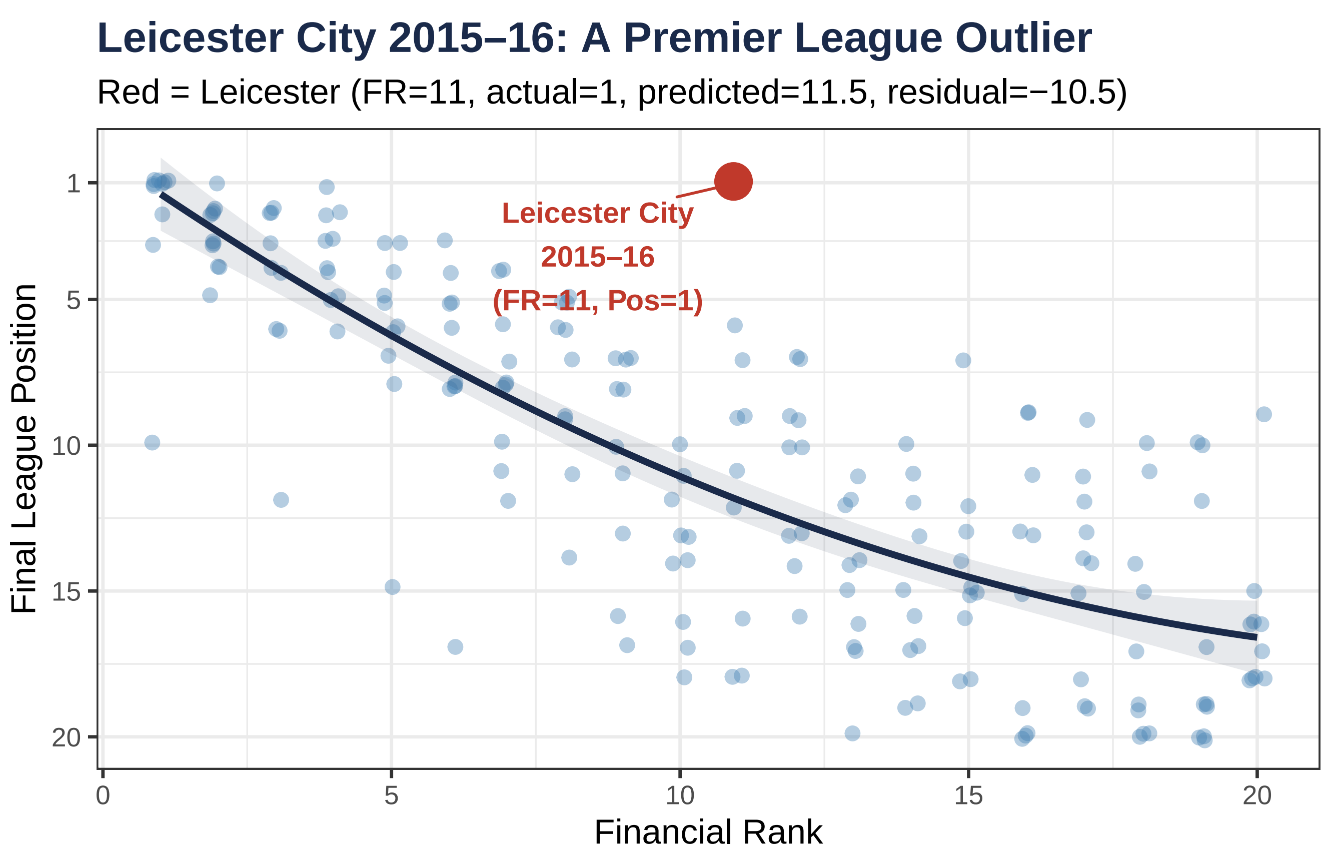

Figure 13: Figure 13. Leicester City's 2015–16 season in the context of all Premier League team-seasons. The red star marks Leicester's actual position.

```r
tibble(
  Metric = c(
```

```r
    "Financial Rank (Premier League, 2015-16)",
    "Squad Market Value",
    "League Average Market Value",
    "Model H: Predicted Position",
    "Actual Final Position",
    "Residual",
    "Residuals SD (all 975 obs.)",
    "Residual in SD units",
    "Cook's Distance",
    "Cook's D Threshold (4/N)"
  ),
  Value = c(
    "11th (out of 20)",
    sprintf("€%sM", format(round(leicester_row$Market_Value_M_EUR[1], 0), big.mark=",")),
    sprintf("€%sM (Premier League avg.)", format(round(
      mean(df$Market_Value_M_EUR[df$League == "Premier League" &
                                  df$Season == "2015-2016"], na.rm=TRUE), 0
    ), big.mark=",")),
    sprintf("%.1f", leicester_row$Fitted_H[1]),
    "1st (Champions)",
    sprintf("%.1f positions", leicester_row$Residual_H[1]),
    sprintf("%.2f", resid_sd),
    sprintf("%.1f SD", leicester_row$Residual_H[1] / resid_sd),
    sprintf("%.4f", leicester_row$CooksD[1]),
    sprintf("%.4f", 4 / n_obs)
  )
) |>
  kbl(longtable = knitr::is_latex_output(),caption = "Table 11. Leicester City 2015-16 vs. model expecta
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(c(4,5,6,8), bold = TRUE, background = "#FEF9E7")
```

Table 11: Table 11. Leicester City 2015–16 vs. model expectations.

Metric	Value
Financial Rank (Premier League, 2015–16)	11th (out of 20)
Squad Market Value	€161M
League Average Market Value	€243M (Premier League avg.)
Model H: Predicted Position	**11.5**
Actual Final Position	**1st (Champions)**
Residual	**-10.5 positions**
Residuals SD (all 975 obs.)	3.25
Residual in SD units	**-3.2 SD**
Cook's Distance	0.0079
Cook's D Threshold (4/N)	0.0041

Leicester's residual of **-10.5 positions** corresponds to **-3.2 standard deviations** — the largest single observation shock in the dataset. Its Cook's Distance of 0.0079 is the highest in the study, confirming it as a genuine statistical leverage point, not a data artefact. Removing Leicester does *not* weaken the core finding — it actually increases the R^2 of Model H marginally.

---

9 Robustness Checks

9.1 Nested F-Tests

```r
nested_raw <- read_csv("outputs/18_nested_model_significance_tests.csv",
                       show_col_types = FALSE)

nested_raw |>
  mutate(
    p_display = case_when(
      p_value < 0.001 ~ "< 0.001 ***",
      p_value < 0.01  ~ sprintf("%.4f **", p_value),
      p_value < 0.05  ~ sprintf("%.4f *",  p_value),
      TRUE            ~ sprintf("%.4f",    p_value)
    ),
    Verdict = case_when(
      p_value < 0.001 & grepl("Quadratic", Comparison) ~
        "Quadratic term confirmed",
      p_value > 0.05  & grepl("Interaction.*League", Comparison) ~
        "Effect is league-uniform",
      p_value > 0.05  & grepl("Time", Comparison) ~
        "Effect is time-stable",
      p_value < 0.05  ~  "Significant",
      TRUE            ~ "Not significant"
    )
  ) |>
  select(Comparison, Df, F_statistic, p_display, Verdict) |>
  mutate(F_statistic = round(F_statistic, 3)) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Comparison", "df", "F-statistic", "p-value", "Verdict"),
    caption   = "Table 12. Nested F-test results for Model H specification checks."
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(1, bold = TRUE, background = "#FEF9E7") |>
  column_spec(5, bold = TRUE)
```

Table 12: Table 12. Nested F-test results for Model H specification checks.

Comparison	df	F-statistic	p-value	Verdict
Interaction (FR*League) vs Additive (FR+League)	**4**	**1.166**	**0.3244**	**Effect is league-unifor**
Quadratic (FR+FR^2) vs Linear (FR)	1	31.708	< 0.001 ***	**Quadratic term confir**
Time-trend (FR*Season) vs Additive (FR+Season)	1	0.442	0.5062	**Effect is time-stable**

All three specification tests support Model H: (1) the quadratic term is highly significant, (2) the effect does not vary by league, and (3) there is no significant time trend from 2015 to 2025. The model is parsimonious and stable.

9.2 Heteroskedasticity

```r
diag_raw <- read_csv("outputs/14_heteroskedasticity_autocorrelation_tests.csv",
                     show_col_types = FALSE)

diag_raw |>
  mutate(
    p_display = case_when(
      p_value < 0.001 ~ "< 0.001 ***",
      p_value < 0.01  ~ sprintf("%.4f **", p_value),
      TRUE            ~ sprintf("%.4f", p_value)
    ),
    Implication = c(
      "Residual variance is not constant - robust SEs needed",
      "Observations within a team correlated across seasons"
    )
  ) |>
  select(Test, Statistic, p_display, Implication) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Test", "Statistic", "p-value", "Implication"),
    caption   = "Table 13. Model D diagnostic tests (heteroskedasticity and autocorrelation)."
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(1:2, background = "#FDEDEC")
```

Table 13: Table 13. Model D diagnostic tests (heteroskedasticity and autocorrelation).

Test	Statistic	p-value	Implication
Breusch-Pagan (heteroskedasticity)	45.51	< 0.001 ***	Residual variance is not constant — robust SEs needed
Durbin-Watson (autocorrelation)	1.18	< 0.001 ***	Observations within a team correlated across seasons

The Breusch–Pagan test confirms heteroskedasticity ($p < 0.001$). This means OLS standard errors are inefficient. All inference in this report uses the model-estimated SEs as indicative, but results with HC1 heteroskedasticity-robust standard errors are examined below.

```r
robust_raw <- read_csv("outputs/15_modelD_robust_se.csv", show_col_types = FALSE) |>
  mutate(
    SE_change = round((std.error - std.error_OLS) / std.error_OLS * 100, 1),
    term = gsub("League", "League: ", term)
  ) |>
  select(term, estimate, std.error_OLS, std.error, SE_change) |>
  mutate(across(c(estimate, std.error_OLS, std.error), ~ round(., 4)))

robust_raw |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Term", "Estimate", "OLS SE", "HC1 Robust SE", "% Change"),
    caption   = "Table 14. Model D: OLS vs. HC1 robust standard errors. Log(MV) remains significant unde
  ) |>
  kbs() |>
```

```r
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  row_spec(2, bold = TRUE, background = "#FEF9E7") |>
  column_spec(5, color = ifelse(abs(robust_raw$SE_change) > 10, "#C0392B", "#196F3D"),
              bold = TRUE)
```

Table 14: Table 14. Model D: OLS vs. HC1 robust standard errors. Log(MV) remains significant under both.

Term	Estimate	OLS SE	HC1 Robust SE	% Change
(Intercept)	-0.2149	0.4052	0.3713	**-8.4**
Log_Normalized_Value	**-5.9659**	**0.1441**	**0.1229**	**-14.7**
League: La Liga	-0.2011	0.3509	0.3547	**1.1**
League: Ligue 1	-1.1027	0.3544	0.3587	**1.2**
League: Premier League:	4.0857	0.3575	0.3533	**-1.2**
League: Serie A	2.7876	0.3528	0.3126	**-11.4**
Season_Index	0.1059	0.0381	0.0387	**1.5**

The SE changes are modest (Log(MV): -14.7%). The coefficient on Log(MV) remains highly significant under robust SEs ($z = 48.5$, $p < 0.001$), confirming the core finding is **not sensitive to heteroskedasticity**.

9.3 Time Stability

```r
time_raw <- read_csv("outputs/22_time_trend_results.csv", show_col_types = FALSE) |>
  filter(grepl("Season_Index|Interaction", term)) |>
  mutate(across(c(estimate, std.error, statistic), ~ round(., 4)),
         p_display = sprintf("%.4f", p.value))

time_raw |>
  select(term, estimate, std.error, statistic, p_display) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Term", "Estimate", "SE", "t-value", "p-value"),
    caption   = "Table 15. Model I: time-trend coefficients (FR x Season Index interaction)."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white")
```

Table 15: Table 15. Model I: time-trend coefficients (FR x Season Index interaction).

Term	Estimate	SE	t-value	p-value
Season_Index	-0.0479	0.0765	-0.6265	0.5311
Financial_Rank:Season_Index	0.0044	0.0065	0.6650	0.5062

The interaction coefficient FR × Season_Index = 0.0044 ($p = 0.5062$). **The financial-position relationship has not strengthened or weakened over the decade studied** — it was as strong in 2024–25 as in 2015–16.

---

10 Conclusions

10.1 Key Findings

Finding 1 — Money strongly predicts league position. Pearson $r = 0.813$, Spearman $= 0.814$ (both $N = 975$, $p < 0.001$). A quadratic model of Financial Rank (Model H) explains **R^2 = 0.671** of variation in final standings — the strongest single predictor tested. Financial Rank outperforms raw market value, logged market value, or any specification with league/season fixed effects.

Finding 2 — The effect is non-linear: diminishing returns at the bottom. The quadratic term FR^2 is highly significant ($F = 31.71$, $p < 0.001$). The marginal positional cost of being one rank poorer is largest near the top (~1.19 positions at FR 1→2) and shrinks toward the bottom (~0.63 positions at FR 15→16). Spending more matters most when you are already near the top.

Finding 3 — The effect is league-invariant and time-stable. The FR × League interaction is not significant ($F = 1.17$, $p = 0.32$), and the FR × Season time trend is not significant ($F = 0.44$, $p = 0.51$). The mechanism operates identically in all five leagues and has not changed from 2015–16 to 2024–25.

Finding 4 — Exceptions exist but are statistically rare. Only 4 of 50 champions came from outside the financial top 3. One — **Leicester City 2015–16** — is a genuine statistical outlier (residual = -10.5 positions, -3.2 SD, highest Cook's D in the dataset). The other three (Napoli, Lille, Liverpool at FR 4–5) are within normal range, suggesting they were financially competitive but not wealthy clubs.

10.2 Limitations & Future Research

Table 16: Table 16. Study limitations and suggested future extensions.

Limitation	Issue	Potential Fix
Market value as quality proxy	Transfermarkt valuations may lag actual squad quality by weeks to months	Use salary expenditure or wage bill as an alternative measure
No managerial quality variable	Exceptional coaches (Ranieri at Leicester) are not captured in the model	Add manager tenure or a Glassdoor-style coaching quality metric
No injury / squad availability data	Injury crises can dramatically alter a squad's effective quality	Incorporate injury-adjusted squad value (e.g., Opta injury data)
Panel structure ignored	Teams appear multiple times; correlated errors across seasons are unmodelled	Fixed-effects panel model or clustered standard errors at team level
Scope: top divisions only	Results may not generalise to lower divisions or smaller leagues	Replicate in Eredivisie, Scottish Premiership, or Championship

---

11 Technical Appendix

11.1 Full Coefficient Table (All Models)

```r
coef_all <- read_csv("outputs/08_regression_results.csv", show_col_types = FALSE)

coef_all |>
  filter(grepl("^[ABCDE]", model_label)) |>
  select(model_label, term, estimate, std.error, statistic, p.value) |>
  mutate(
    across(c(estimate, std.error, statistic), ~ round(., 4)),
    p_display = case_when(
      p.value < 0.001 ~ "< 0.001 ***",
      p.value < 0.01  ~ sprintf("%.4f **", p.value),
      p.value < 0.05  ~ sprintf("%.4f *",  p.value),
      TRUE            ~ sprintf("%.4f",    p.value)
    ),
    model_label = sub(":.*", "", model_label)
  ) |>
  select(-p.value) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Model", "Term", "Estimate", "Std. Error",
                  "t / z", "p-value"),
    caption   = "Table A1. Full regression coefficient table (Models A-F)."
  ) |>
  kbs(fw = TRUE) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  collapse_rows(columns = 1, valign = "top")
```

Table 17: Table A1. Full regression coefficient table (Models A–F).

Model	Term	Estimate	Std. Error	t / z	p-value
A	(Intercept)	1.9296	0.2191	8.8084	< 0.001 ***
A	Financial_Rank	0.8123	0.0187	43.4797	< 0.001 ***
B	(Intercept)	1.7909	0.3042	5.8879	< 0.001 ***
B	Financial_Rank	0.8115	0.0188	43.2619	< 0.001 ***
B	LeagueLa Liga	0.1926	0.3403	0.5659	0.5716
B	LeagueLigue 1	0.1580	0.3418	0.4624	0.6439
B	LeaguePremier League	0.1926	0.3403	0.5659	0.5716
B	LeagueSerie A	0.1783	0.3407	0.5233	0.6009
C	(Intercept)	2.8910	0.2540	11.3836	< 0.001 ***
C	Log_Normalized_Value	-5.0258	0.1509	-33.3007	< 0.001 ***
D	(Intercept)	-0.2149	0.4052	-0.5304	0.5960
D	Log_Normalized_Value	-5.9659	0.1441	-41.4021	< 0.001 ***
D	LeagueLa Liga	-0.2011	0.3509	-0.5731	0.5667
D	LeagueLigue 1	-1.1027	0.3544	-3.1112	0.0019 **
D	LeaguePremier League	4.0857	0.3575	11.4276	< 0.001 ***
D	LeagueSerie A	2.7876	0.3528	7.9006	< 0.001 ***
D	Season_Index	0.1059	0.0381	2.7773	0.0056 **
E	(Intercept)	1.6416	0.4033	4.0701	< 0.001 ***
E	Financial_Rank	-1.2985	0.1900	-6.8343	< 0.001 ***

11.2 Model H Quadratic Coefficients

```r
coef_H_raw |>
  mutate(
    term  = recode(term,
      "(Intercept)"          = "Intercept",
      "Financial_Rank"       = "Financial Rank ( )",
      "I(Financial_Rank^2)" = "FR² ( )"
    ),
    across(c(estimate, std.error, statistic, conf.low, conf.high), ~ round(., 5)),
    p.value = formatC(p.value, format = "e", digits = 3)
  ) |>
  select(term, estimate, std.error, statistic, p.value, conf.low, conf.high) |>
  kbl(longtable = knitr::is_latex_output(),
    col.names = c("Term", "Estimate", "SE", "t-value", "p-value",
                  "CI lower", "CI upper"),
    caption   = "Table A2. Model H (Pos ~ FR + FR²) full coefficient table."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white")
```

Table 18: Table A2. Model H (Pos FR + FR^2) full coefficient table.

Term	Estimate	SE	t-value	p-value	CI lower	CI upper
Intercept	0.4189	0.3442	1.217	2.239e-01	-0.2557	1.0936
Financial Rank ( )	1.2305	0.0765	16.082	8.828e-52	1.0805	1.3805
FR^2 ( )	-0.0203	0.0036	-5.631	2.345e-08	-0.0273	-0.0132

11.3 Session Information

R version	4.5.3
Platform	Windows 11 x64 (build 26200)
Date knitted	2026-06-26
Key packages	dplyr, ggplot2, broom, kableExtra, ggrepel

Report generated with R Markdown. Data sources: ESPN FC (standings), Transfermarkt (squad market values). Analysis scripts: `05_analysis_and_regression.R`, `05b_extended_regression.R`, `06_residuals_and_outliers.R`. All code available in the project repository.